

Recent Progresses on Defect Passivation toward Efficient Perovskite Solar Cells

Feng Gao, Yang Zhao, Xingwang Zhang, and Jingbi You*

The disorderly distribution of defects in the perovskite or at the grain boundaries, surfaces, and interfaces, which seriously affect carrier transport through the formation of nonradiative recombination centers, hinders the further improvement on the power conversion efficiency (PCE) of perovskite solar cells (PSCs). Several defect passivation strategies have been confirmed as an efficient approach for promoting the performance of PSCs. Herein, recent progress in the defect passivation toward efficient perovskite solar cells are summarized, and a classification of common passivation strategies that elaborate the mechanism according to the location of the defects and the type of passivation agent is presented. Finally, this review offers likely prospects for future trends in the development of passivation strategies.

1. Introduction

In less than a decade, perovskite solar cells (PSCs) have made tremendous progress in photoelectric conversion efficiency (PCE), rising from 3.8% in 2009 to 25.2% in 2019.^[1,2] This remarkable progress is due to its superior optoelectronic properties: excellent light absorption coefficient ($\approx 10^5 \text{ cm}^{-1}$), long carrier diffusion lengths in films ($>1 \mu\text{m}$) and high defects tolerance. Furthermore, based on its simple solution preparation method, the leveled cost of electricity (LCOE) of perovskite photovoltaic modules was calculated to be 3.5–4.9 US cents kWh^{-1} (efficiency: 12%/lifetime: 15 years),^[3] which is almost less than half the LCOE of crystalline silicon solar cells.^[4,5]

In the field of photovoltaic technologies, the golden triangle elements, the efficiency, cost and lifetime (or stability), are often used to judge the commercial feasibility of solar cell modules.^[6] For the PSCs, efficiency and stability are the core issues of concern among the three elements. In view of the simple processability of perovskite halides, one could expect non-negligible defects at grain boundary and interface caused

by low temperature processing. The presence of these defects leads to a decrease in crystal quality, and the energy band misalignment between the active layer with respect to the charge transport layer, thus seriously affecting charge carrier transportation, which could be one of the most important reasons for approaching its theoretical efficiency limit of 30.5%.^[7]

In addition to efficiency, another factor that needs serious consideration is device stability. Compared to the typical lifetime of silicon solar cell of 25 years, the longest lifetime expectancy reported for PSC is only 10 000 h, about 1 year.^[8] There are many external conditions that affect the

stability of the perovskite device, which can be summarized into the following four categories: water, oxygen, heat, and light.^[9] Thereinto, water and oxygen stability can be solved by external encapsulation, while the photo- and thermal-stability are fully dependent on the characteristics of the material itself: thermal instability and ion migration. The mobile ions accrued at the perovskite interface rebound to chemically react with adjacent contact sites, which can be completely expedited by photons and heat.^[10,11]

Passivation, defined as a material becoming “passive,” that is, being less affected by the external environment, is an effective strategy to reduce perovskite defects and suppress ion movement. As early as the beginning of the silicon solar cell industry in the 1990s, dielectric coatings, like silicon nitride, silicon dioxide or titanium dioxide, have already been the universal solution to surface passivation. Surface passivation also plays a vital role in thin film solar cells such as cadmium telluride (CdTe) and $\text{CuIn}_x\text{Ga}_{(1-x)}\text{Se}_2$ (CIGS).^[12] Like these thin films, perovskite also contains many intrinsic and interface defects. Defects make perovskites active and prone to decomposition, leading to the formation of nonradiative recombination, which in turn affects device performance and stability. According to previous reports, the following types of defects in perovskite materials have a huge impact on device performance: 1) intrinsic 0D defect, including vacancy, interstitial and antisite substitution;^[13–20] 2) 0D defects caused by impurities, such as doped or permeable ions; 3) 2D defects, like grain boundaries and surface defects.^[21] All of these defects will affect carrier recombination, charge transport, and band alignment.^[22]

In this review, we sum up the achievements of enhancing PCE of PSCs by defect passivation in the last few years. The different passivation strategies for perovskite layer and also the charge transport layer are systematically summarized. A simple

F. Gao, Y. Zhao, Prof. X. Zhang, Prof. J. You
Key Laboratory of Semiconductor Materials Science
Institute of Semiconductors
Chinese Academy of Sciences
Beijing 100083, P. R. China
E-mail: jyou@semi.ac.cn

F. Gao, Y. Zhao, Prof. X. Zhang, Prof. J. You
Center of Materials Science and Optoelectronics Engineering
University of Chinese Academy of Sciences
Beijing 100049, P. R. China

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outlook on the possible problems and future development trend of the passivation technology is also expounded.

2. The Effect of Defects on Performance of Perovskite Solar Cells

The working principle of perovskite solar cells is shown in **Figure 1**, which includes the entire physical process of charge transfer. When the solar cell is under operation, the incident photons ($h\nu > E_g$) are absorbed by the material. The electrons are excited from valence band into conduction band, then photo generated electron-hole pairs are separated into electrons and holes by built-in electric field. After diffusion, transport and extraction processes, electrons and holes are collected by the negative and positive electrodes respectively.^[23] The above process is based on the ideal state. In fact, the charge will undergo another process—recombination, including the Auger recombination, the Shockley–Read–Hall (SRH) recombination, surface and interface recombination, all of which are nonradiative recombinations. The open-circuit voltage (V_{OC}) is produced by the tearing of the quasi-Fermi level of electrons and holes,^[24] and the extra charge path caused by the nonradiative recombination is the fundamental source of the open-circuit loss.^[25,26]

Effective carrier lifetime (τ_{eff}) is a physical parameter commonly used to characterize crystal quality, which can be defined as $\tau_{eff} = \Delta n / U_{eff}$ (Δn is excess-charge carrier time, U_{eff} is the effective recombination rate). Considering the main types of recombination in perovskites, τ_{eff} can be characterized as follows^[27]

$$\frac{1}{\tau_e} = \frac{1}{\tau_{bulk}} + \frac{1}{\tau_{surf}} = \frac{1}{\tau_{Aug}} + \frac{1}{\tau_{SRH}} + \frac{1}{\tau_{surf}} \quad (1)$$

In this expression, the bulk carrier time τ_{bulk} is related to the Shockley–Read–Hall recombination (τ_{SRH} , induced by defects in the bulk perovskite) and the Auger recombination (τ_{Aug}),^[28,29] and τ_{surf} stands for the recombination induced by surface and interface.^[30] Thus, we dig out the theoretical connection between defects and recombination.

In perovskite, usually taking MAPbI₃ as an example, MA⁺ (CH₃NH₃⁺) is a monovalent cation, Pb²⁺ is a bivalent metal cation, and I[−] is a halogen anion. MA⁺ can be replaced by FA⁺ (NH₂–CH=NH₂⁺) or Cs⁺, Pb²⁺ can be changed to Sn²⁺, and I[−]



Feng Gao received his B.S. degree in new energy science and engineering at School of Renewable Energy of North China Electric Power University. He is currently a graduate student under the supervision of Prof. Jingbi You at Institute of Semiconductors, Chinese Academy of Sciences. His research interests focus on the design of high-performance and stable perovskite solar cells.



Jingbi You is currently a full professor in the Institute of Semiconductors, Chinese Academy of Sciences (ISCAS). He received his Ph.D. degree in material sciences from ISCAS in 2010, and later, he did his postdoc at the University of California, Los Angeles, from 2010 to 2015, mainly working in organic tandem solar cells and perovskite solar cells. Since 2015, he joined ISCAS as a full professor. His present research interests are organic/inorganic semiconductor materials and their optoelectronic devices such as solar cells, LEDs, and detectors.

can be substituted for other halogens.^[31,32] Therefore, 0D point defects often found in perovskites include Pb²⁺ vacancies (V_{Pb}), I[−] vacancies (V_I), interstitial I₃[−], interstitial Pb²⁺, and Pb–I anti-site (Pb_I , I_{Pb}),^[33] which are illustrated in **Figure 2**. Point defects are the recombination centers for trapping electrons or holes, thereby correlating with τ_{SRH} . As we all know, some of the point defects with high formation energies give rise to deep levels in the bandgap, and the more important ones that would produce

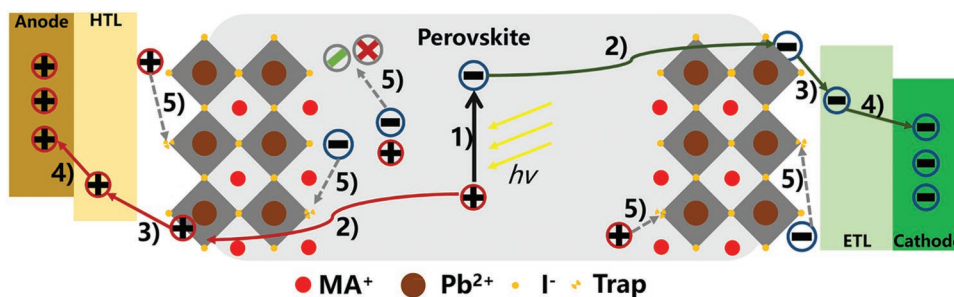


Figure 1. Working principle of perovskite solar cells: 1) Charge dissociation, 2) charge diffusion, 3) charge transport, 4) charge extraction, and 5) charge recombination.

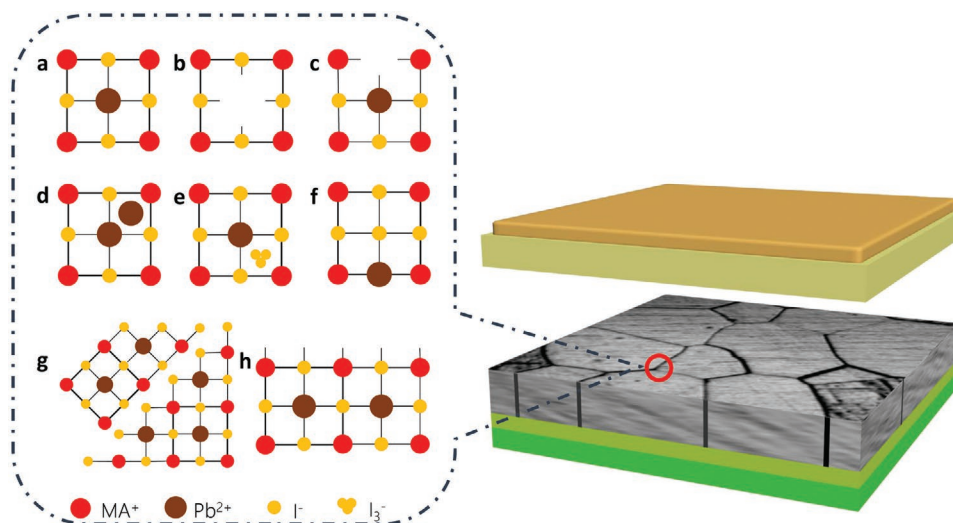


Figure 2. Schematic illustration of typical defects in PSCs: a) perfect lattice, b) Pb^{2+} vacancies, c) I^- vacancies, d) interstitial Pb^{2+} , e) interstitial I_3^- , f) Pb–I antisite substitution, g) grain boundary, and h) dangling bonds on the surface.

shallow levels have low formation energies. More research focuses on grain boundaries and surfaces, where the crystal structure is suddenly interrupted, producing a large number of localized surface states. These states are related to tension, usually described by dangling bonds.^[34] Drawing on the passivation strategy in traditional solar cells, 2D perovskites, insulating polymers and other materials with wide bandgaps are often used to passivate such defects in PSCs.

Furthermore, defects exert an influence on other photovoltaic parameters of PSCs. Defects inside the transport layer and poor contact between the transport layer and the electrodes affect charge extraction, resulting in series resistance, which reduces the J_{SC} and FF (fill factor).^[35] Moreover, severe defects such as micro-cracks and scratches on the surface cause rough surface topography, in turn affecting FF. More importantly, it is considered that the movement of iodide ions is a vital source of the instability and hysteresis.^[17,36–38] Thus, the parameters in PSCs including J_{SC} , FF, and especially V_{OC} , could be improved by passivation.

Despite great efforts, the specific role of defects in perovskites is still vague. Characterization techniques like photoluminescence (PL),^[39–41] time-correlated single photon counting (TCSPC),^[42] Kelvin probe force microscopy (KPFM) and thermal admittance spectroscopy (TAS) are often used to delineate the effect of defect passivation.^[43,44] Some new tools, like electron-beam-induced current (EBIC),^[45] secondary ion mass spectrometry (SIMS) and scanning tunneling microscope (STM)^[46,47] are utilized to explore the location and nature of defects, supplemented by density functional theory (DFT) calculation.^[48,49] We fully believe that a better understanding of the changes in the nature of defects in passivation will be obtained with the help of new characterizations, thereby further optimizing the passivation strategy.

3. Passivation of Perovskite Layer

According to the location of the defects, we will elaborate the passivation mechanism of different types of passivators for

bulk defects and grain boundary surface defects in perovskite respectively. The effect of different passivation strategies will be summarized in Table 1.

3.1. Passivation of Perovskite Bulk

The bulk defects in perovskite are mainly point defects, including various ion vacancies, interstitial and substitutional ions, which cause nonradiative recombination to affect device performance. It is widely accepted that the trap levels of most point defects are close to the valence band or conduction band.^[115,116] Therefore, the focus of the passivation is the point defects with shallow energy level.

3.1.1. Passivation by Incorporation of Metal Cations

It is a common method of introducing a specific ion to dope ionic compounds with a certain proportion in perovskite precursor solutions. The introduction of metal cations usually has a positive effect on the I interstitial and Pb–I antisite substitution.

Since Cs^+ is a suitable A-site ion of a reasonable radius,^[55] alkali metal ions are often used as dopant. That PV performance enhancement of $\text{Cu}_2\text{ZnSn}(\text{S},\text{Se})_4$ by alkali ion doping further motivating researchers to focus on it.^[117] As early as 2017, Bi et al. revealed for the first time that Na^+ has a unique effect on the performance of PSCs when storage in N_2 atmosphere.^[47] By SIMS measurements, they found that the residual Na in the ITO glass will diffuse to the device after long-term storage (Figure 3a). It is expected that Na^+ ions contribute to the PCE improvement by a passivation effect on the grain boundaries (GBs) in perovskite.

Subsequently, Son et al. systematically studied the passivation effect of all alkali metal ions.^[50] As shown in Figure 3c, they doped 10 μmol of alkali metal iodide (LiI, NaI, KI, RbI, or CsI) in perovskite precursor solution, and hysteresis nearly

Table 1. Summary of passivation strategies for perovskite layer.

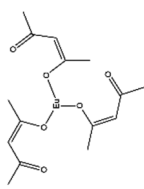
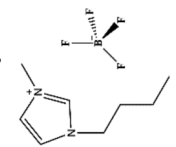
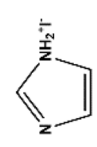
Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P^a)	J_{sc} [mA cm ⁻²] (C/P^a)	FF [%] (C/P^a)	PCE [%] (C/P^a)	Ref.
Na ⁺	Na ⁺	CsMAPbI ₃	Na ⁺	1.05/1.06	23.5/24.4	76.3/78	18.8/20.2	[47]
KI	KI	FA _{0.85} MA _{0.15} PbI _{2.55} Br _{0.45}	K ⁺	1.114/1.128	22.04/21.47	69.8/72.5	17.14/17.55	[50]
KI	KI	CsMAFAPb(I/Br) ₃	KBr	1.05/1.17	22.6/23.2	73/79	17.3/21.5	[51]
KI	KI	Cs _{0.06} FA _{0.79} MA _{0.15} Pb(I _{0.85} Br _{0.15}) ₃	K ⁺	1.07/1.18	22.65/23.06	75/79	18.2/21.6	[52]
RbI	RbI	Cs _{0.06} FA _{0.79} MA _{0.15} Pb(I _{0.85} Br _{0.15}) ₃	Rb ⁺	1.09/1.14	22.58/23.11	74/78	18.1/20.6	[52]
KI	KI	CsMAFAPb(I/Br) ₃	KBr	1.09/1.13	22.77/22.88	77/78	19.26/20.25	[53]
RbI	RbI	CsPbI ₂ Br	Rb ⁺	1.20/1.24	13.86/14.02	66/69	11.02/12	[54]
RbI	RbI	CsPbI ₂ Br	Rb ⁺	1.20/1.24	13.86/14.02	66/69	11.02/12	[54]
Cesium acetate (CsAc)	CH ₃ COOCs	FA _{0.85} MA _{0.15} PbI ₃	Cs ⁺ , Ac ⁻	1.08/1.12	24.35/24.92	73.85/78.64	19.47/21.95	[55]
NiCl ₂	NiCl ₂	MAPbI ₃	Ni ²⁺	1.08/1.13	21.55/22.64	74/81	17.25/20.6	[56]
Europium(III) acetylacetonate		MAPbI ₃	Eu ³⁺ -Eu ²⁺	1.15 ^{b)}	23.53	80.7	21.89	[57]
ZnI ₂	ZnI ₂	CsPbI ₂ Br	Zn ²⁺	1.123/1.236	13.27/15.61	68.5/63	10.21/12.16	[58]
GeI ₂	GeI ₂	CsSn _{0.5} Ge _{0.5} I ₃	GeO ₂	0.63 ^{b)}	18.41	61.3	7.11	[59]
MAI	MAI	FAPbI ₃	Cl ⁻ , MA ⁺	1.03/1.13	24.84/25.92	77.08/82.02	19.66/24.02	[60]
NaF	NaF	CsFAMA _{0.98} Br _{0.02}	F ⁻ , hydrogen bonds (N—H...F)	1.095/1.126	24.03/24.23	74.77/80.35	19.68/21.92	[48]
NbF ₅	NbF ₅	FA _{0.85} MA _{0.15} PbI ₃	Nb ⁵⁺ , F ⁻	1.03/1.05	24.45/25.40	72.56/77.08	18.30/20.56	[61]
BMIMBF ₄		(FA _{0.83} MA _{0.17}) _{0.95} Cs _{0.05} Pb(I _{0.9} Br _{0.1}) ₃	BMIM ⁺ , BF ₄ ⁻	1.02/1.08	23.2/23.8	79/81	18.7/20	[62]
Iodine	I ₂	(FAPbI ₃) _{0.87} (MAPbBr ₃) _{0.13}	I ₂	1.116/1.19	22.5/23.2	78.2/78.2	20.49/21.65	[63]
PbI ₂	PbI ₂	FAMAPb(I/Br) ₃	PbI ₂	1.12 ^{b)}	23.86	80.6	21.64	[64]
PbI ₂	PbI ₂	FAMAPb(I/Br) ₃	PbI ₂	1.04/1.15	23.8/24.8	75.2/78.4	18.8/22.3	[65]
MAI	CH ₃ NH ₃ I	MAPbI ₃	I ⁻ , MA ⁺	1.03/1.12	23.9/23.4	68/76	16.8/19.8	[66]
EAI	C ₂ H ₅ NH ₃ I [†]	FA _{0.9} Cs _{0.07} MA _{0.03} Pb(I _{0.92} Br _{0.08}) ₃	EA ⁺	1.07/1.12	24.38/24.36	77.8/80.4	20.72/22.3	[67]
IAI		FA _{0.9} Cs _{0.07} MA _{0.03} Pb(I _{0.92} Br _{0.08}) ₃	IA ⁺	1.07/1.10	24.38/24.14	77.8/79.4	20.72/21.6	[67]

Table 1. Continued.

Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^a	J_{sc} [mA cm ⁻²] (C/P) ^a	FF [%] (C/P) ^a	PCE [%] (C/P) ^a	Ref.
GuaI		FA _{0.9} CS _{0.07} MA _{0.03} Pb(I _{0.92} Br _{0.08}) ₃	Gua ⁺	1.07/1.10	24.38/24.45	77.8/75.3	20.72/20.9	[67]
BAI		FA _{0.83} CS _{0.17} Pb(I _{0.6} Br _{0.4}) ₃	BA ⁺	1.14/1.18	19.8/19.8	75/73	16.9/17.2	[68]
EDA ₂		FASnI ₃	EDA ²⁺	0.36/0.58	17.6/21.3	63/72	4.0/8.9	[69]
FAI + iso-BAI		(FAPbI ₃) _{0.85} (MAPbI ₃) _{0.15}	BA ⁺	1.09/1.17	22.9/23.1	78/82	19/21.7	[70]
Octylammonium iodide		MAPbI ₃	OA ⁺	1.06/1.11	21.8/22.6	79/82	18.4/20.6	[71]
Octane-1,8-diammonium iodide(C8)		MAPbI ₃	-NH ₂	1.09/1.11	17.9/19.4	75/79	14.7/17	[72]
FACI + ADAHCl		(FAPbI ₃) _{0.87} (MAPbI ₃) _{0.13}	Cl ⁻	1.13/1.16	23.0/23.5	78/78.5	19.4/21.2	[73]
PEAI		FA _{0.92} MA _{0.08} PbI ₃	PEA ⁺ (aromatic ring and -NH ₂), I ⁻	1.12/1.16	24.24/24.96	78/81	21.2/23.5	[39]
Benzylamine		FACsMAPb(I _{0.87} Br _{0.13}) ₃	Aromatic ring, -NH ₂	1.01/1.08	19.39/21.37	66.8/73.9	13.06/17.05	[74]
FAL		FAPbI ₃	Para-fluorine substituent, aromatic ring, -NH ₂	0.96/1.03	22.3/23.3	65/64	14.2/17.3	[75]
IPFB		CS _{0.17} FA _{0.83} PbI _{2.6} Br _{0.4}	Fluorine substituent, aromatic ring	1.02/1.08	22.05/22.79	71/75.6	15.97/18.60	[76]
Adamantine (ADA)		MAPb(I _{0.87} Br _{0.13}) ₃	ADA	1.02/1.06	23.8/23.38	57/67	13.0/15.7	[77]
1-Adamantylamine		CsMAFAPb(I _{0.87} Br _{0.13}) ₃	ADA, -NH ₂	1.08/1.14	22.47/22.53	79.2/79.3	19.32/20.47	[78]
C ₆₀		CsMAFAPb(I _{0.87} Br _{0.13}) ₃	Fullerene	1.08/1.16	22.47/22.57	79.2/80.4	19.32/20.93	[78]
		MAPbI ₃		0.89/0.92	18.85/21.07	80/80	13.37/15.44	[40]

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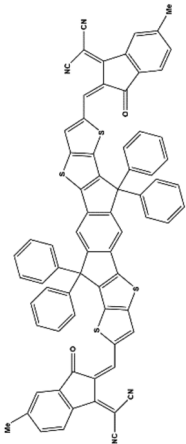
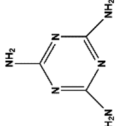
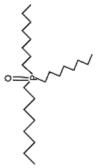
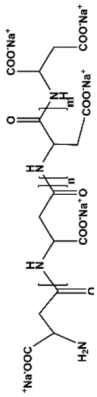
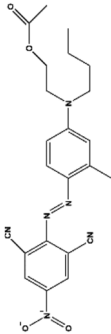
Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^a	J_{sc} [mA cm ⁻²] (C/P) ^a	FF [%] (C/P) ^a	PCE [%] (C/P) ^a	Ref.
IT-M		C _{50.05} (MA _{0.1} FA _{0.83}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃	Thiophene, -C=O	1.11/1.14	22.2/22.34	75.6/77.7	18.63/19.70	[79]
Thiophene		MAPb ₂ (Cl) ₃	S donor	0.95/1.02	20.7/21.3	68/68	12.1/14.3	[41]
Pyridine		MAPb ₂ (Cl) ₃	N donor	0.95/1.05	20.7/24.1	68/72	12.1/15.5	[41]
Imidazolium iodide		MAPbI ₃	N donor	1.06/1.11	22.18/22.88	77.5/79.3	18.22/20.13	[80]
Melaminium iodide		(FAPbI ₃) _{0.875} (CsPbI ₃) _{0.125}	N donor	1.075/1.082	21.66/21.80	77.4/76.4	18.02/18.43	[81]
Triethyl phosphine oxide (TOPO)		MAPbI ₃	O donor	1.1/1.1	19.7/19.7	79/77	17.2/16.7	[82]
Polyspartic acid sodium (PASP)		MAPbI ₃	Thiophene, -C=O	1.01/1.07	21.48/22.8	79.17/82.2	17.18/20.05	[83]
Ethyl cellulose		MAPbI ₃	O donor, hydrogen bonds	1.05/1.1	22.77/22.89	71/77	17.11/19.41	[84]
AVA1		MAPbI ₃	AVA ⁺ (-NH ₂ , O donor)	1.05/1.02	21.45/18.84	70.3/75.5	15.95/14.6	[8]
2-MP		MAPbI ₃	S, N donor	1.09/1.16	22.56/22.61	74.64/77.44	18.35/20.28	[85]

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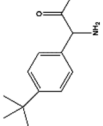
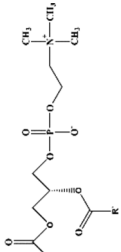
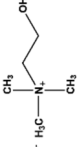
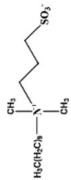
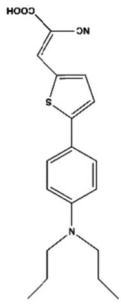
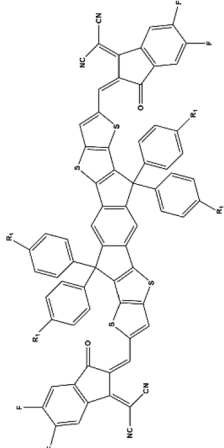
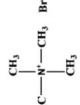
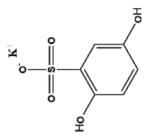
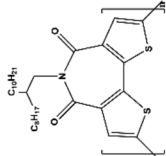
Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^a	J_{sc} [mA cm ⁻²] (C/P) ^a	FF [%] (C/P) ^a	PCE [%] (C/P) ^a	Ref.
D4TBP		Cs _{0.05} FA _{0.81} MA _{0.14} Pb _{1.25} Br _{0.45}	—NH ₂ , O donor	1.08/1.16	22.4/22.51	78.8/80.3	19.1/21.0	[86]
L-α-Phosphatidylcholine		MAPbI ₃	—N(CH ₃) ₃ ⁺ , —PO ₄ ⁻	1.04/1.08	22.5/22.7	73/80	17.1/19.6	[87]
Choline chloride		FA _{0.85} MA _{0.15} PbI _{2.55} Br _{0.45}	—N(CH ₃) ₃ ⁺ , Cl ⁻	1.03/1.14	23.7/23.7	79/78	19.2/21	[87]
DPSI		FA _{0.85} MA _{0.15} PbI _{2.55} Br _{0.45}	—N(CH ₃) ₃ ⁺ , —SO ₃ ⁻	1.06/1.13	22.7/23.3	80/80	19.1/21.1	[88]
SP3		MAPbI ₃	—COO—	1.07/1.15	22.5/22.69	77.2/78.55	18.52/20.43	[89]
IT-4F		MAPbI ₃	—C=O, CN, thiophene	1.06/1.12	20.79/21.35	70.74/76.74	15.59/18.35	[90]
Me ₄ NBr		(FASnI ₃) _{0.6} (MAPbI ₃) _{0.4}	—N(CH ₃) ₃ ⁺	0.64/0.76	25.68/26.33	70.57/72.81	11.81/14.56	[91]
KHQSA		FASnI ₃	—SO ₃ ⁻	0.36/0.55	17.26/17.64	56.3/69.4	3.53/6.76	[92]
PBTI		CsMAFAPb ₂ (IBr) ₃	O, S donor	1.07/1.13	22.72/22.91	77.7/79.8	18.89/20.67	[93]

Table 1. Continued.

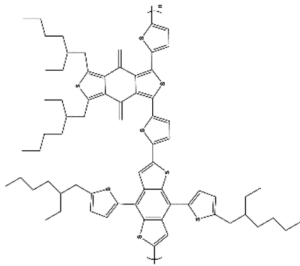
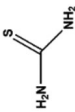
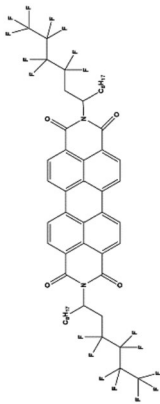
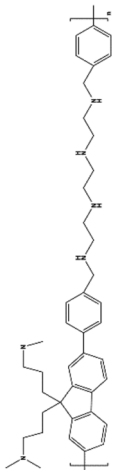
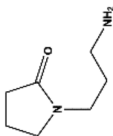
Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^a	J_{sc} [mA cm ⁻²] (C/P) ^a	FF [%] (C/P) ^a	PCE [%] (C/P) ^a	Ref.
PBDB-T		Cs _{0.04} MA _{0.14} FA _{0.82} Pb(I _{0.88} Br _{0.14}) ₃	Thiophene	1.075/1.113	21.73/22.39	74/77.8	17.28/19.38	[94]
Thiourea		MA _{0.9} FA _{0.1} PbI ₃	S donor	1.05/1.08	23.27/23.53	74/77	18.08/19.25	[95]
F-PDI		Cs _{0.05} (MA _{0.17} FA _{0.83}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃	Fluorine substituent	1.06/1.065	21.24/22.74	78.25/79.90	17.62/19.26	[96]
PN4N\ PDCBT		CsPbI ₂ Br	N, O, S donor	1.08/1.30	14.8/15.3	76.7/81.5	12.3/16.2	[97]
N-(3-Aminopropyl)-2-pyrrolidinone (NAP)		MASr _{0.5} Pb _{0.5} I _{3-x} Cl _{3-x}	O donor, -NH ₂	0.66/0.71	23.3/27.1	67.1/69.5	10.3/13.4	[98]

Table 1. Continued.

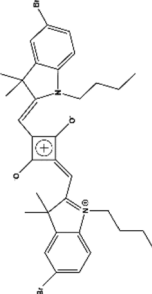
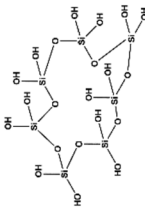
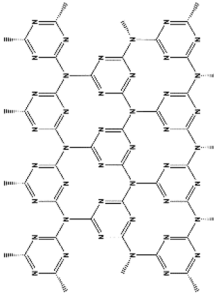
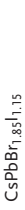
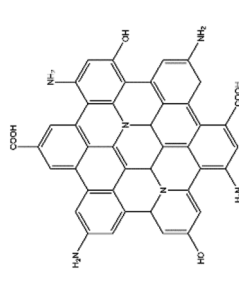
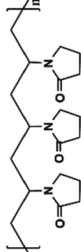
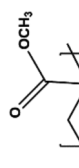
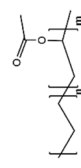
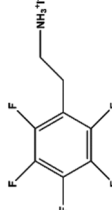
Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^a	J_{sc} [mA cm^{-2}] (C/P) ^a	FF [%] (C/P) ^a	PCE [%] (C/P) ^a	Ref.
SQ63 (squaraine)		MAPbI ₃	Four-membered ring, O donor	1.03/1.07	21.04/22.18	71.64/79.03	15.77/18.83	[99]
OS		MAPbI ₃	SiO ₂	1.10/1.15	22.5/22.7	77.1/80.9	19.1/21.1	[100]
$\text{g-C}_3\text{N}_4$		MAPbI ₃	$\text{g-C}_3\text{N}_4$	1.08/1.16	21.7/23.0	77/79	18.0/21.1	[101]
HTAB		(FAPbI ₃) _{0.95} (MAPbBr ₃) _{0.05}	$-\text{N}(\text{CH}_3)_3^+$	0.93/1.15	24.79/24.92	69/79.9	15.8/22.8	[103]
MAPbBr ₃		MAPbI ₃	MAPbBr ₃	0.94/1.05	19.28/21.25	62.7/74.5	12.05/16.58	[104]
CsPbBr ₃		Cs _{0.05} FA _{0.80} MA _{0.15} Pb _{1.55} Br _{0.45}	CsPbBr ₃	1.16/1.195	22.7/22.5	76.4/77	20.1/20.6	[105]
CsPbBr _{1.85} I _{1.15}		Cs _{0.05} (MA _{0.15} FA _{0.85}) _{0.95} Pb(_{0.85} Br _{0.15}) ₃	CsPbBr _{1.85} I _{1.15}	1.09/1.14	23.37/23.35	76/79	19.51/21.14	[105]
CQDs		MAPbI ₃	$-\text{NH}_2$, O donor	1.10/1.13	21.88/22.19	72.9/74.75	17.59/18.81	[106]
C ₆ Br		(FAPbI ₃) _{0.92} (MAPbBr ₃) _{0.08}	$-\text{NH}_2$, Br ⁻	1.14/1.16	24/23.9	78.2/80	21.3/22.3	[107]
C ₈ Br		(FAPbI ₃) _{0.92} (MAPbBr ₃) _{0.08}	$-\text{NH}_2$, Br ⁻	1.14/1.19	24/24.2	78.2/78.5	21.3/22.6	[107]

Table 1. Continued.

Passivator	Structure	Perovskite components	Passivation functional groups	V_{oc} [V] (C/P) ^{a)}	J_{sc} [mA cm ⁻²] (C/P) ^{a)}	FF [%] (C/P) ^{a)}	PCE [%] (C/P) ^{a)}	Ref.
Polystyrene (PS)		MAPbI ₃	Aromatic ring	1.07/1.10	21.1/22.9	74.9/80.6	16.9/20.3	[108]
PEO		CsMAFAPb(I/Br) ₃		1.14/1.16	21.86/22.32	77.3/79.2	19.24/20.46	[109]
PVP		MAPbI ₃	—C=O	1.05/1.10	19.65/19.87	74/75	15.29/16.46	[110]
PMMA		CsPbI ₃	—N—C=O	1.11b)	14.88	65	10.74	[111]
EVA		Cs _{0.07} Rb _{0.03} FA _{0.08} MA _{0.135} PbI _{2.55} Br _{0.45}	—C=O	1.10/1.21	22.85/22.60	77/76.4	18.50/20.80	[112]
FEAI		Cs _{0.1} MA _{0.2} FA _{0.7} PbI _{2.55} Br _{0.45}	—C=O	1.04/1.06	21.78/22.39	78/81	17.69/19.31	[113]
		FA _{0.92} MA _{0.04} Cs _{0.04} PbI ₃	FEA ⁺	1.05/1.1	25.47/25.79	77.5/78.4	20.62/22.16	[114]

^{a)} C/P represents the parameters of control/passivated device; ^{b)} if there is only one value, it represents the device parameters after passivation.

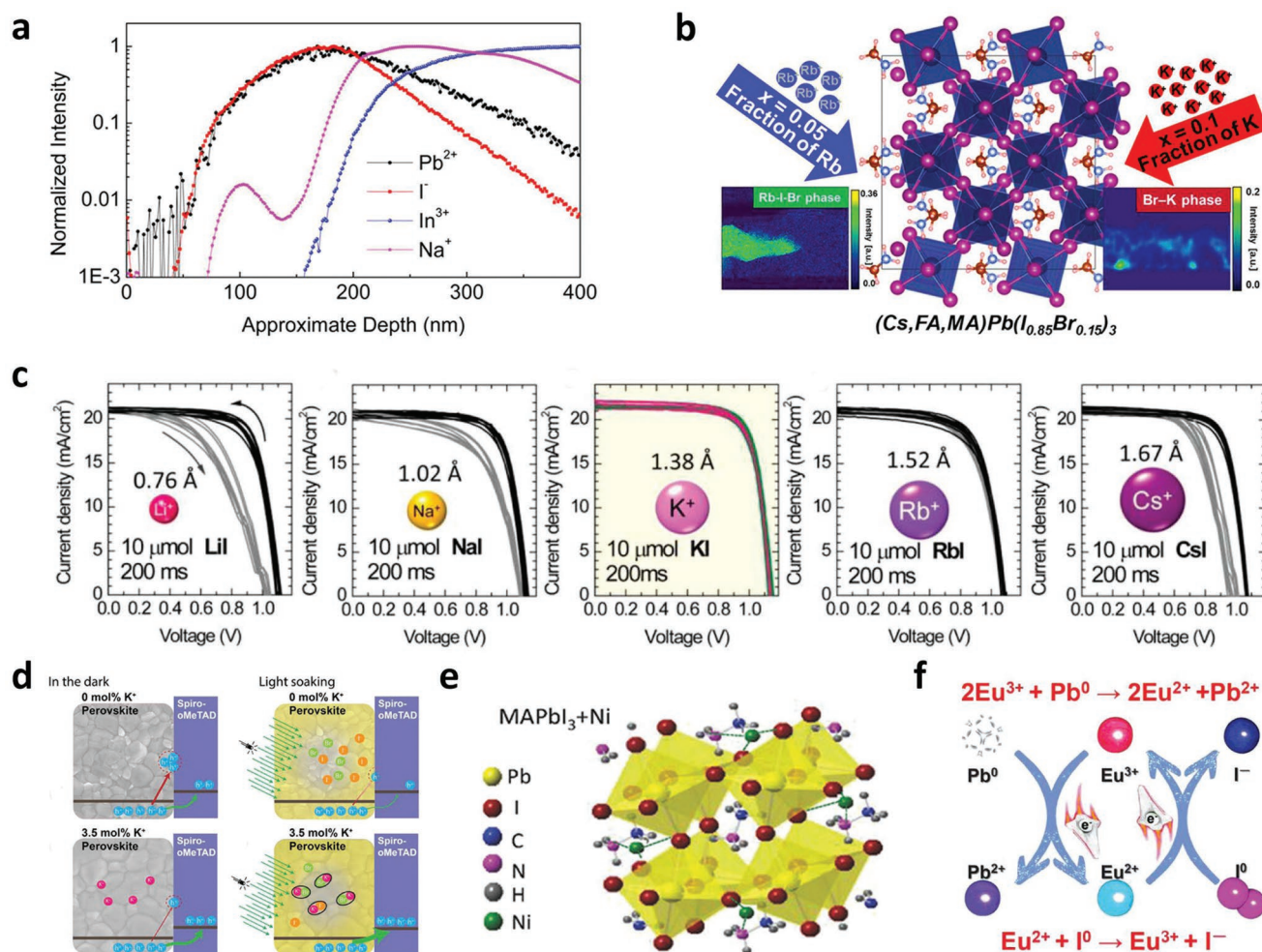


Figure 3. Passivation by metal ions. a) SIMS measurement of MAPbI₃ films grown on an ITO glass substrate. Reproduced with permission.^[47] Copyright 2017, American Chemical Society. b) Non-negative matrix factorization (NMF) decomposition results of K-passivated and the Rb-passivated sample. Reproduced with permission.^[52] Copyright 2018, American Chemical Society. c) J-V curves of perovskite solar cells based on perovskite doped with 10 μmol of LiI, NaI, KI, RbI, and CsI. Reproduced with permission.^[50] Copyright 2018, American Chemical Society. d) Schematic of the passivation effect of K⁺-doped perovskite triggered by illumination: 0 mol% K⁺ perovskite and 3.5 mol% K⁺ perovskite in the dark and during light soaking, respectively. Reproduced with permission.^[53] Copyright 2019, Wiley-VCH. e) The crystal structure of MAPbI₃(Ni) perovskite, in which Ni-I and Ni-N bonds are indicated with green lines. Reproduced with permission.^[56] Copyright 2018, Wiley-VCH. f) Proposed mechanism diagram of cyclically elimination of Eu³⁺-Eu²⁺ metal ion pair. a.u., arbitrary units. Reproduced with permission.^[57] Copyright 2019, American Association for the Advancement of Science.

disappears only when doped by KI. They added KI in different compositions of perovskite precursor solutions, and similar results were obtained, indicating that the passivation effect of K⁺ has certain universality. Due to the lower formation energy of the K-position in the mixed cation system, the effect of eliminating hysteresis is better. Additionally, what is interesting to mention is that the V_{OC} values for the forward scan are increased and almost coincides with the reverse scan curve after KI doping.

Rb⁺ ions have also been studied separately. Abdi-Jalebi et al. indicated that the distribution of I is relatively homogeneous all over the film, while K and Br are distributed with a certain proportion at the grain boundary (Figure 3b).^[51] K and Br form superior compounds, which immobilize the surplus halide and the inhibit halide migration.^[118] However, the passivation effect of Rb is far less than K, and the explanation et al. is that

the solubility of nonperovskite phase in the Rb doped device (Rb-I-Br-based phase) is worse than the K doped one (KBr), which has negative effects on optoelectronic performance.^[52] Besides, moisture could promote grain coalescence in K-passivated device, so the humidity stability of the device has been significantly improved. Zheng et al. pointed out another key element of K passivation: light soaking (Figure 3d).^[53] Light soaking induces the formation of KBr-like compounds at the grain boundaries, and the device preserves the excellent passivation effect even after being placed in the dark once triggered by light. It has also been reported that Rb⁺ has a certain passivation effect on all inorganic PSCs (CsPbI₂Br).^[54] Here we can conclude that alkali metal ions, especially K⁺ doping are effective passivators to improve battery performance.

Divalent or trivalent metal cations are also commonly used as passivators. Wang et al. introduced Ni²⁺ ions into the PbI₂

precursor solution, which mainly distributed among MA cations and octahedral $[\text{PbI}_6]^{4-}$ and had strong impact on suppress generation of the Pb interstitial and Pb–I antisite substitution (Figure 3e).^[56] With an optimized addition of Ni^{2+} , device efficiency was greatly improved from 17.25% to 20.61%. In 2018, Wang et al. studied the effect of rare earth element doping on the performance of PSCs.^[57] After comparing the three ions of Eu^{3+} , Y^{3+} , and Fe^{3+} , they found that Eu^{3+} had a positive effect on both the PCE and the stability of device. By utilizing high-resolution X-ray photoelectron spectroscopy (XPS), they examined the ratio of I/Pb, and considered that doped Eu form an ion pair $\text{Eu}^{3+}\text{--Eu}^{2+}$, acting as a “redox shuttle” that selectively oxidized Pb and reduced I interstitials (Figure 3f). The devices retained 90% of the initial PCE after 8000 h storage with Eu incorporated, which showed good light stability at the same time, suggesting Eu ion is an excellent passivator. However, not all passivation by metal ion doping are through ionic bonding. Chen et al. reported that the passivation mechanism of doping Ge^{2+} in Sn-based perovskite cells is due to the formation of a native oxide layer on the surface.^[59] The role of other metal cations in passivation is listed in Table 1. Apart from metal cations, organic cations possessing passivation effects, especially those with ammonium radicals also attracted a lot of attentions. The mechanism of that will be further discussed in Section 3.2.2.

3.1.2. Passivation by Incorporation of Anions

Contrary to cations, anions are often incorporated to passivate Pb interstitial and halide vacancies. There are relatively few researches about anion passivation, which usually focus on halide ions. It is generally believed that excessive iodide ions in the precursor solution or iodine added separately have an inhibitory effect on halogen vacancies.^[119,63] There are also reports on the passivation of other small radius halogen ions. Thanks to the appropriate radius, bromine ions can replace iodine in $[\text{PbI}_6]^-$ octahedron to adjust the bandgap and benefit the film formation.^[51,120–122] Kim et al. believed that Br^- promoted the separation of electrons and holes at grain boundaries.^[123] Another common halogen ion, Chloride ions are often doped by adding PbCl_2 , MACl , and FACl to the perovskite precursor.^[124,125,73] Although Cl ions are believed to volatilize during annealing,^[126] studies have shown that residual Cl will passivate grain boundaries and surfaces.^[127] Recently, Kim et al. achieved amazing PCE of 23.48% in the pure FAPbI_3 PSCs by adding MACl .^[60] They revealed that Cl^- stabilized the perovskite structure thermodynamically with lower formation energy and promoted the formation of high quality crystal α -phase FAPbI_3 .

As the smallest radius of the halogen ion, the fluoride ion has the strongest electronegativity and can form a strong chemical bond with Pb. By comparison with device modified by adding 0.1% NaX ($\text{X} = \text{I}, \text{Br}, \text{or Cl}$) and DFT calculations, Li et al. argued that the F^- had a strong passivation effect on Pb interstitial and halide vacancies by forming robust ionic bonds with lead and hydrogen bonds ($\text{N--H}\cdots\text{F}$) with MA^+/FA^+ cations.^[48] In addition, F^- ions may accumulate on the surface of the perovskite and passivate surface defects. Bai et al. introduced the ionic liquid BMIMBF₄ (containing four BF_4^- groups)

into the perovskite film, markedly improving the long-term stability of device.^[62] Abate et al. illustrated the passivation of halide ions defects by iodopentafluorobenzene through supramolecular fluoride bond complexation.^[77]

Except for halogen anions, there are other anions that also can be passivators.^[61,128,129] It has been reported that halogen-like ions SCN^- can reduce grain boundary recombination through reaction intermediates.^[129] Yin et al. found that O^{2-} had an effective effect on grain boundary passivation and advance charge transfer.^[128]

3.2. Grain Boundaries and Surface Defects Passivation

For the perovskite polycrystalline film prepared by spin coating, the grain boundary and surface defects cannot be eliminated no matter what kind of crystallization method and device structure are used. Similar to silicon-based solar cells, dangling bonds generated by discontinuous crystal structures at the surface can cause serious recombination of carriers, greatly reducing device performance. Even though perovskite has a high tolerance to bulk defects, effective surface passivation is still one of the cores for pushing the PCE of PSCs.

3.2.1. Native Passivation by PbI_2

As a product of perovskite decomposition after crystallization, PbI_2 aroused the attention of researchers as early as 2013.^[130] Since Wang et al. first verification of the existence of PbI_2 through femtosecond time-resolved transient absorption spectroscopy technology (fs-TA), a series of targeted research work have been carried out.^[131] It usually appears at the edge of the grain and can be clearly observed by scanning electron microscopy (Figure 4e) and scanning kelvin probe microscopy (Figure 4d,f).^[64,44] As shown in Figure 4a, Chen et al. gave a reasonable speculation about the passivation mechanism by using I-type band alignment principle. The bandgap of PbI_2 is 2.3 eV, which is larger than that of perovskite (1.5 eV). And the conduction band as well as the valence band of PbI_2 are located above and below the perovskite respectively, so the carrier transportation is blocked in the grain boundary enriched in PbI_2 . In addition, PbI_2 change the bowing direction of the grain to grain boundary bending, thereby reducing the carrier recombination at the interface (Figure 4b).

You et al. fabricated planar-structure PSCs with efficiency beyond 21% with PbI_2 passivation in 2017.^[64] They are convinced that the amount of PbI_2 has a significant impact on device performance, and excessive PbI_2 may cause severe hysteresis and poor stability in the device. When PbI_2 is over excess, the hole extraction rate of at the interface is faster than that of electrons, causing electron accumulation and hysteresis. Meanwhile, insufficient organic component (FA/MA) may contribute to ion migration and affect device stability. With the help of light-modulated scanning tunneling microscopy (LMSTM), another research group gave a point that it could enhance charge separation and reduce back recombination when the thickness of the PbI_2 passivation layer was designed to be less than 20 nm.^[132]

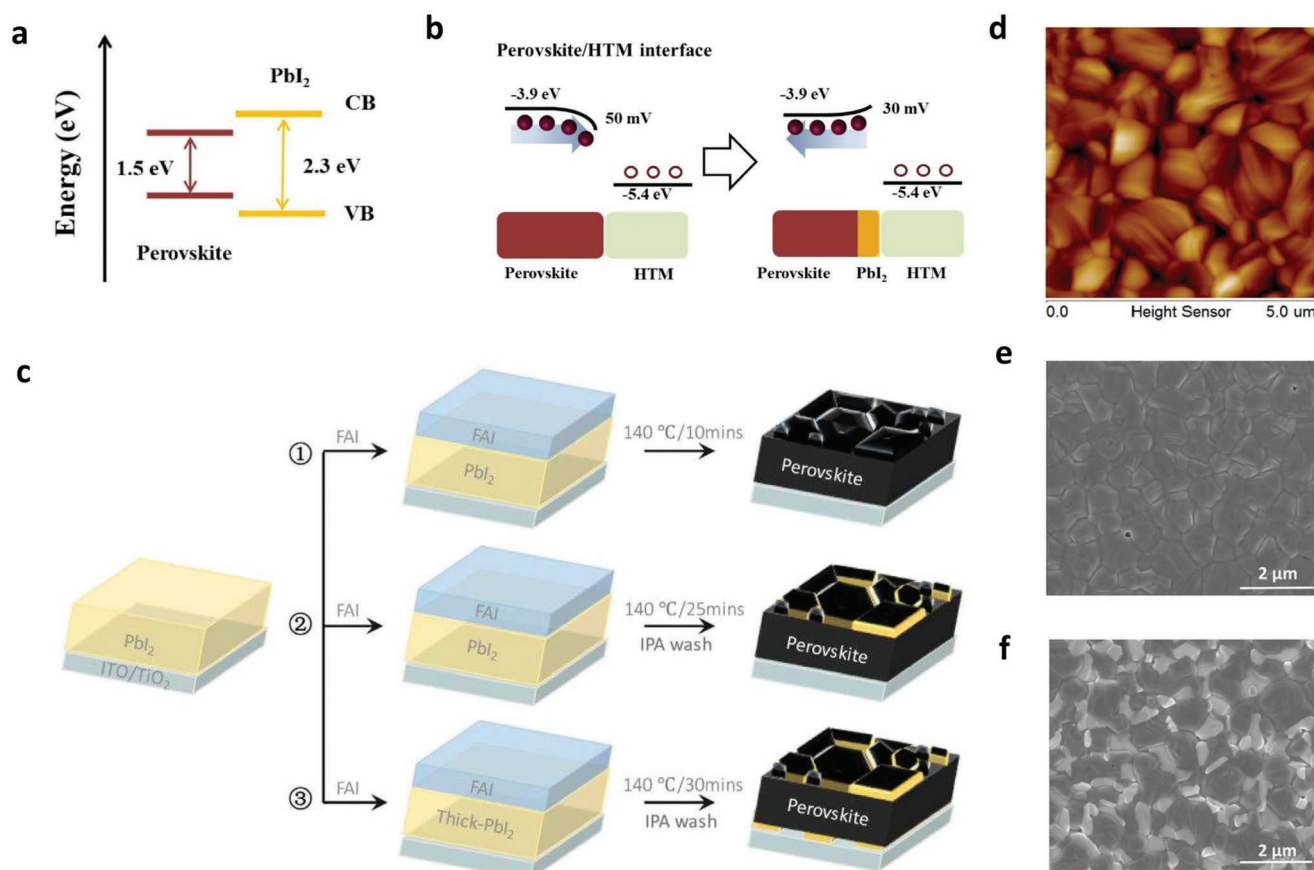


Figure 4. Native passivation by PbI_2 : a) the coexistence of PbI_2 and perovskite in the film shows a type I alignment of the band edge, with the bandgap of 2.3 and 1.5 eV, respectively.^[44] b) The interface of perovskite/HTM is changed by PbI_2 . Reproduced with permission.^[44] Copyright 2014, American Chemical Society. c) Schematics for fabrication procedures of perovskite films with varying PbI_2 distribution at front side, rear side, and grain boundaries. Reproduced with permission.^[65] Copyright 2018, Wiley-VCH. Scanning electron microscopy (SEM) images of perovskite films with different PbI_2 /FAI combinations, e) no PbI_2 was observed on the surface, f) observable white phase of PbI_2 formed on the perovskite surface. Reproduced with permission.^[64] Copyright 2017, Wiley-VCH.

Similar to the native passivation of SiO_2 in silicon-based solar cells,^[133] PbI_2 is also considered to have a very broad passivation effect on perovskites. Zhao et al. fabricated double-side-passivated PSCs (stabilized efficiency of 22%) with PbI_2 distributed to both the front/rear-side surfaces.^[65] As illustrated in Figure 4c, they managed the residual amount of PbI_2 by controlling the annealing time and the thickness of the PbI_2 layer. The under layer does not participate in the reaction during annealing (with sufficient thickness of PbI_2), resulting in residues on the back side of the device.

3.2.2. Passivation by Alkylammonium Halogen

As one of the raw materials for the synthesis of perovskite films, alkylammonium halogen salts are a popular addition material that can improve device performance by doping or layering. This type of material always has a comprehensive passivation effect on both the cation and anion defects in the bulk or on the surface, and may further enhance the passivation effects through hydrogen bonding or ionic bonding, thus being one of the best passivators for high-efficiency perovskite devices.

Firstly, we focus our attention on the linear alkylammonium halogen. In 2016, Son et al. discovered that excess methylammonium iodide (MAI) in perovskite precursor solution may self-form a thin layer near the grain boundaries by STEM (Figure 5a), which significantly reduced device hysteresis.^[66] Afterward, Qi et al. used XPS and UPS to further confirm that not the MAI but the dissociated species adjusted the energy level of the interface between perovskite and hole-transport layer.^[134] It has also been reported that MAI steam treatment could diffuse MA^+ ions into the perovskite layer and combine with water molecules to act as a hydrophobic agent.^[135]

Other short linear alkaneamine functionalities, including ethylammonium (EA^+),^[67] *n*-butylammonium (BA^+),^[68] isobutylammonium (iso-BA^+) and guanidinium (Gua^+) mostly improve the device performance and lifetime by forming a layer of low-dimensional perovskite on the surface.^[67,136] Wang et al. realized that BA^+ constituted a 2D layered perovskite with a wide bandgap on a 3D perovskite.^[68] When carriers transport to the grain boundary in the 3D perovskite layer, they are blocked by the 2D-3D heterojunction and continue to propagate in the grain without generating recombination (Figure 5c). Alharbi et al. analyzed the modification of the perovskite surface via the

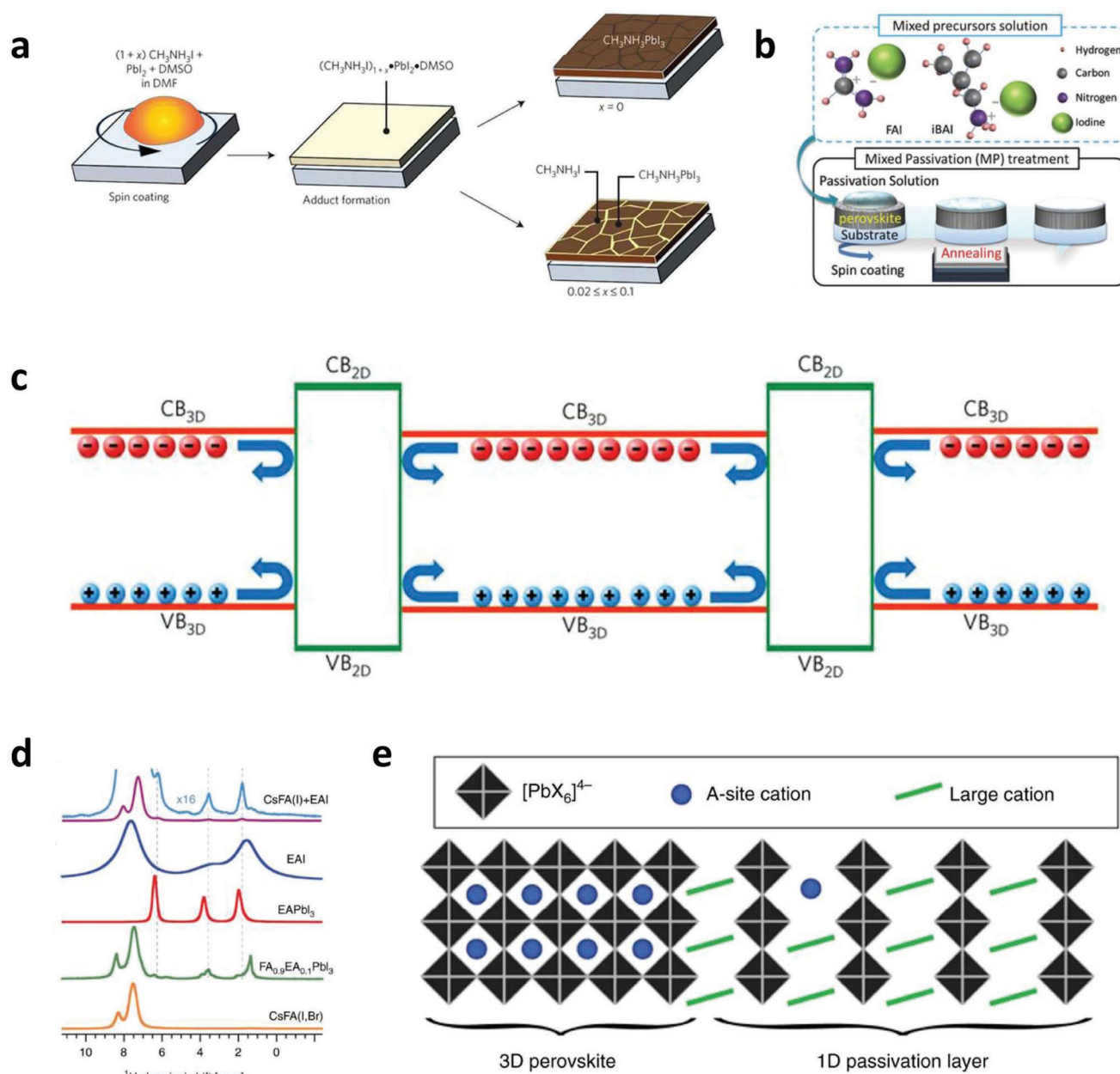


Figure 5. Passivation by linear alkylammonium halogen. a) Schematic representation of the coating process used to form the $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite films with and without excess MAI. Reproduced with permission.^[66] Copyright 2016, Macmillan. b) Schematic of the MP passivation treatment method with FAI and iso-BAI. Reproduced with permission.^[70] Copyright 2018, Wiley-VCH. c) Proposed electronic band offsets of the 2D–3D heterojunction. CB and VB stand for conduction band and valence band, respectively, with the subscript indicating either the 2D or 3D phase. Reproduced with permission.^[68] Copyright 2017, Macmillan. d) Solid-state ^1H MAS NMR measurements at 21.1 T, 300 K, and 20 kHz MAS (unless noted otherwise) of thin film of CsFA(I) treated with 5 mg mL^{−1} EAI (60 kHz MAS), neat EAI, bulk mechanochemical EAPbI₃, bulk mechanochemical FA_{0.9}EA_{0.1}PbI₃, and thin film of FA_{0.93}Cs_{0.07}Pb(I_{0.92}Br_{0.08})₃.^[67] e) Schematic representation of the 1D/3D heterostructure evidenced by solid-state NMR proximity measurements. Reproduced with permission.^[67] Copyright 2019, Macmillan.

addition of EA⁺ and Gua⁺.^[67] The results of solid-state magic angle spinning (MAS) NMR (Figure 5d) showed that EAI fully converted into new mixed EA/FA phases and formed a 1D perovskite passivation layer without annealing (Figure 5e). Ethylenediammonium diiodide (EDAI₂) and BAI have been shown to have a similar effect in Sn-based perovskites.^[69] Cho et al. developed a mixed passivation (MP) treatment for perovskite by

using FAI and iso-BAI to control the morphology of the film (Figure 5b), getting a stabilized power conversion efficiency of 21.7%.^[70]

On the contrary, long linear alkaneamine radicals, such as octylammonium (OA⁺) and $\text{NH}_3\text{I}(\text{CH}_2)_8\text{NH}_3\text{I}$ (C₈),^[71,72] do not form a 2D perovskite (because of the high formation) but directly act as passivator itself as reported. However,

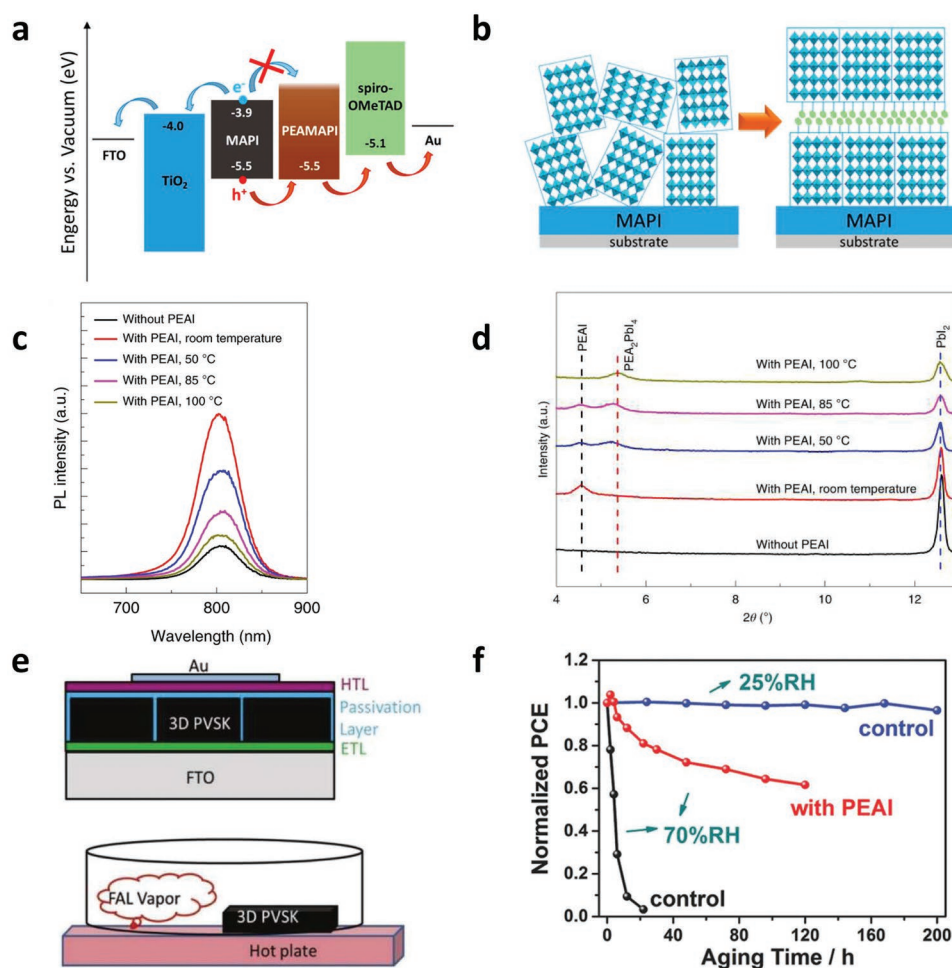


Figure 6. Passivation by alkylammonium halogen with benzene. a) Schematic energy level diagram of a MAPbI₃/PEAMAPbI₃ heterojunction solar cell.^[138] b) Schematic interpretation of the surface reorganization of MAPbI₃ films upon PEAMAPbI₃ formation. Reproduced with permission.^[138] Copyright 2016, American Chemical Society. c) Steady PL of the perovskite films with PEAI treatment under different conditions.^[39] d) XRD of perovskite films with PEAI treatment under different conditions. Reproduced with permission.^[39] Copyright 2019, Macmillan. e) Schematic illustration of the expected perovskite solar cell with ultrathin passivation layer, and nondestructive passivation process by vapor post-treatment. Reproduced with permission.^[76] Copyright 2018, Wiley-VCH. f) The moisture stability test of inverted devices without PEAI treatment. Reproduced with permission.^[74] Copyright 2017, Wiley-VCH.

n-hexylammonium bromide (C₆Br) is an exception, and a device with PCE of 22.6% and electroluminescence external quantum efficiency up to 8.9% has been prepared by passivation using this material to form layered perovskite.^[107] Another special class of alkylammonium salts is adamantylamine, whose extraordinary spatial structure can also abate the surface dangling bonds of perovskite.^[78,137]

Next, let us look at the alkylammonium halogen with benzene, among which the passivation effect of phenethylammonium iodide (PEAI) is the most eye-catching. In 2016, Hu et al. first introduced PEA into the PSCs by spin-coating PEA/MAI dissolved in IPA.^[138] With the help of grazing-incidence wide-angle X-ray scattering (GIWAXS), they observed the formation of the perovskite bilayer. As shown in Figure 6a, the increase in open-circuit voltage of the device is due to the layered perovskite hindering electron transport, reducing interface recombination and promoting hole extraction. They also pointed out that the reason why the passivation effect of PEA was better

than BAI might be that the 2D passivation layer formed by PEA is more orderly and facilitates the longitudinal transport of carriers (Figure 6b).

Especially, You et al. spotted that PEA could play a better role of passivation without annealing according to PL results (Figure 6c).^[39] The PCE of the device prepared by this method is as high as 23.32%, which was once a world record. By comparing the XRD results of PEA at different annealing temperatures (Figure 6d), they judged that PEA would not form 2D perovskite without annealing. The surprising passivation effect of PEA stems from the synergy of the three groups: the π -conjugation structure of the benzene ring promotes charge transport and reduces the neutral iodine related defects; the amine group forms a hydrogen bond with the iodide ion and coordinate with the Pb²⁺ interstitials; the iodide ions fill the iodine vacancies.

By conducting a controlled trial of aniline, benzylamine, and phenethylamine, Wang et al. indicated that devices' better

moisture resistance was owing to the hydrophobicity of the phenyl group (Figure 6f).^[75,74] However, the insulation of PEA and the possible deterioration of light absorption make the J_{SC} and FF of the device worse.^[139] Besides, Zhao et al. grew a 4-fluoroaniline (FAL) passivation layer by vapor post-treatment (Figure 6e), the conjugated amine in aromatic ring and *para*-fluorine substituent facilitate both defect passivation and charge transfer.^[76] Overall, alkylammonium halogen salts are near-perfect passivation agents based on their powerful integrated passivation effect.

3.2.3. Passivation by Lewis acid and Base

Lewis acids and bases are a class of substances that have a particular effect on specific defects. A Lewis acid refers to a molecule or ion that accepts a foreign electron pair, while a Lewis base represents a molecule or ion that gives an electron pair. Such molecules could combine with covalent coordinate bonds to form acid–base complexes without electron transfer. Consequently, Lewis acids have unique passivation capabilities for defects with lone pairs of electrons like free iodide ions and lead iodine antisite defects; electron-deficient defects such as Pb^{2+} interstitials can be strongly bonded by Lewis bases (Figure 7a).^[140] Deep-level trap composite centers caused by defects of both anion and cation can be well eliminated, and the overall performance of the device will be significantly improved.

The most representative Lewis acid passivators are fullerene (C_{60}) and its derivatives. They are also good electron transport materials because of their inherent electron attraction in fullerene spherical structures.^[19,141] In 2014, Shao et al. found that thin layers of PCBM could eliminate hysteresis and reduce traps caused by I^- anions defects.^[43] Furthermore, Liang et al. compared the passivation effect of three fullerene derivatives (C_{60} , PCBM, and ICBA) and determined C_{60} to stand out according to the results of PL intensity (Figure 7b).^[40]

The most prominent feature of Lewis bases is the presence of atoms containing lone pairs of electrons, especially nitrogen,^[41,80,81] sulfur,^[142,95] oxygen, and phosphorus atoms.^[82,83,94] Derivatives of pyridine and thiophene are the first Lewis bases to be used to passivate perovskites.^[41] Noel et al. achieved a 3% efficiency gain through treatment of the perovskite layer surfaces with the Lewis bases thiophene and pyridine (Figure 7c).^[41] The increase in PL lifetime (Figure 7d) close to an order of magnitude demonstrated that thiophene and pyridine have good passivation effects on defects, which stems from the covalent bond of Pb^{2+} to this species. Zhang et al. developed efficient and stable PSCs by doping imidazolium iodide and fumigating with methylamine.^[80] This process deprotonated the imidazolium cations and released the imidazole, where cation exchange improved film quality. Kim et al. reported a hysteresis-free PSC with melaminium iodide additive, and the hysteresis index decreased from 0.1282 to 0.0674 (Figure 7g).^[81]

Regarding the Lewis base containing O and P elements, deQuilettes et al. investigated three Lewis bases, including trioctylphosphine oxide (TOPO), octa-decanethiol, and triphenylphosphine,^[82] and showed that TOPO is the passivator

(Figure 7e,f). Slightly different, graphene and ethyl cellulose (EC) also apply hydrogen bonds to passivate defects in addition to Lewis bases,^[143,84] and hydroxybenzene sulfonic acid is believed to improve the oxidation resistance of tin-based PSCs.^[92]

Molecules with polyfunctional groups have synergistic and more powerful passivation effects than monofunctional molecules. The combination of a carboxyl or carbonyl group and an amine group is a common assembly. A molecule with these two functional groups will have multiple pairs of lone pairs of electrons, and the ability to capture defects is enhanced. Aminovaleric acid (AVA), containing both a carboxyl group and an amine group, has proven to be effective in improving device humidity stability.^[8,144] Whether directly doped or layered separately, AVA always appears at the surface and grain boundaries to terminate surface dangling bonds (Figure 8a).^[144] Besides, some articles pointed out that AVA formed a layer of 2D perovskite on the surface.^[8]

As shown in Figure 8b, Zhang et al. designed a new molecule 2-mercaptopyridine (2-MP) including both sulfhydryl and pyridine to bidentate anchor the Pb^{2+} cation more strongly (compared with pyridine and *p*-toluenethiol).^[85] The strong bonding between the passivating agent and the lead ion makes it impossible for the external water molecules to contact the perovskite layer, which greatly enhances the humidity stability (Figure 8d). The PCE of pure MAPbI₃ perovskite devices boosted from 18.35% to 20.38%, with a V_{OC} of 1.18 V. After comparing a series of molecules with carboxyl group and amine group, Yang et al. synthesized a new molecule *D*-4-*tert*-butylphenylalanine (D4TBP) with 4-*tert*-butylphenyl, amine, and carboxyl.^[86] With the combination of multiple functional groups, this small molecule has a very significant passivation effect on Lewis acid defects (Figure 8c). The open-circuit voltage loss of the device prepared in this way is only 0.34 V, which is the lowest reported in the literature.

Another valuable idea is to add Lewis acid and Lewis base, and the resulting zwitterion molecules have a comprehensive passivation effect on positive and negative charge defects. Zheng et al. designed a series of choline molecules based on this principle, in which 1- α -phosphatidylcholine contained the quaternary ammonium group ($-N(CH_3)_3^+$) and phosphate group ($-PO_4^-$) while choline chloride had the negatively charged chloride ion.^[87] The common passivation effect of the two types of defects, anionic Pb–I antisites and cationic Pb^{2+} , makes the PCE of PSC reach 21% with an open-circuit loss of only 0.39 V (Figure 8f). After that, they designed another molecule naming 3-(decyldimethylammonio)-propane-sulfonate inner salt (DPSI), in which the anion was replaced with sulfonic group ($-SO_3^-$).^[88] The molecule plays dual roles in adjusting film growth, reducing pinhole formation and passivating multiple defects, ultimately resulting in a device PCE of up to 21.1%.

Wu et al. ingeniously applied organic donor– π –acceptor (D– π –A) molecules with different electron density distributions to the passivation project, and the excellent passivation for lead ion defects of them increased the V_{OC} by 0.08 V (Figure 8e).^[89] There are many other passivation cases that use similar principles, such as IT-4F,^[90] Me₄NBr and the alkylammonium halogen salt mentioned in Section 3.2.2.^[91] Integrated passivation

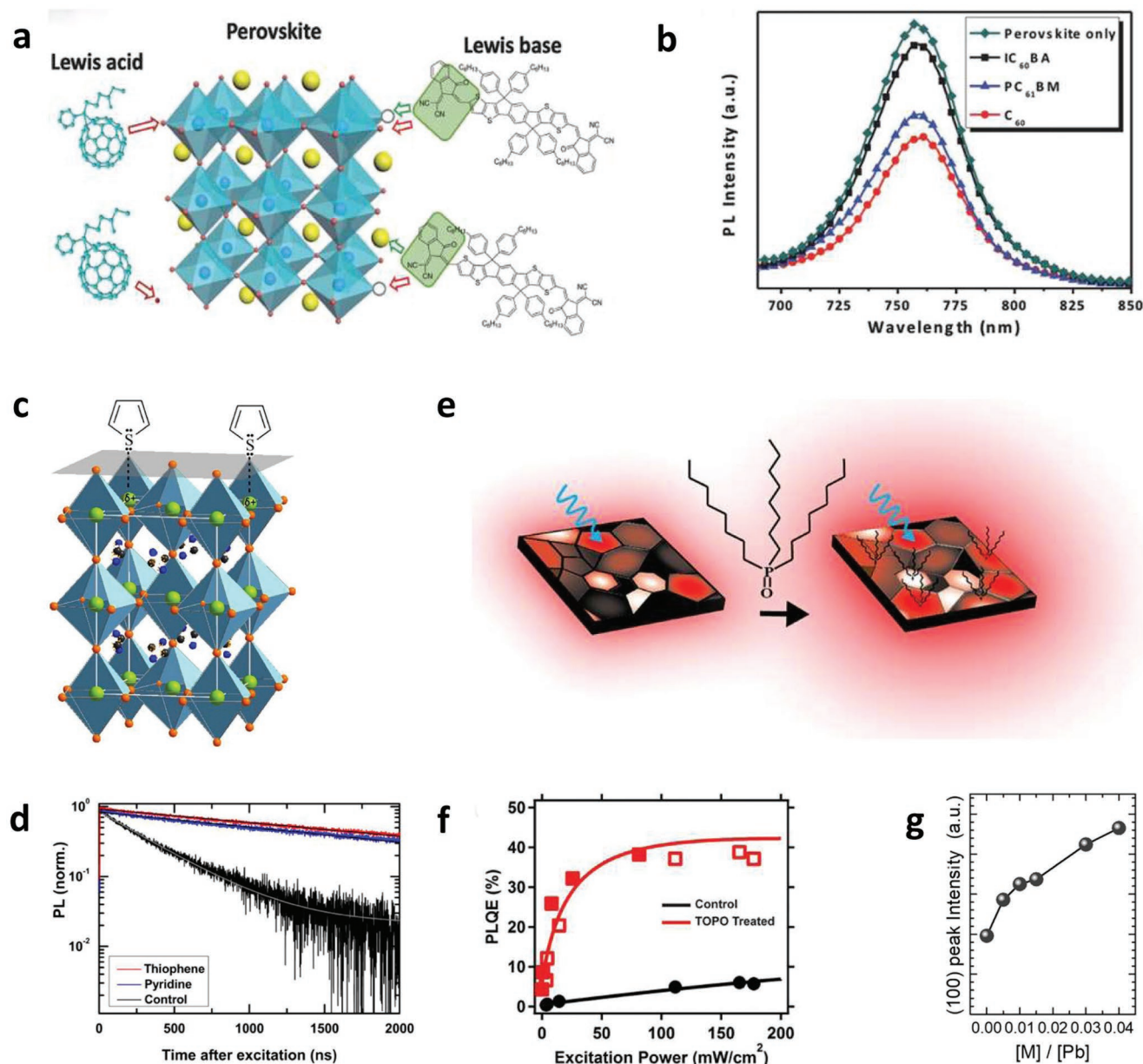


Figure 7. Passivation by a monofunctional Lewis base. a) Selective interactions between Lewis acid or base functional groups and the perovskite crystal facets. Reproduced with permission.^[140] Copyright 2018, Wiley-VCH. b) Steady-state PL spectra of perovskite in the presence of the studied fullerene quenchers. Reproduced with permission.^[40] Copyright 2015, Wiley-VCH. c) Proposed passivation mechanism of thiophene or pyridine.^[41] d) Photoluminescence characteristics of bare and thiophene or pyridine passivated films. Time-resolved PL measurements of control, thiophene, and pyridine samples with 507 nm pulsed (200 kHz) excitation at a range of pulse fluences corresponding to the quoted photoexcitation densities. Reproduced with permission.^[41] Copyright 2014, American Chemical Society. e) Schematic diagram of TOPO for PLQY enhancement.^[82] f) External PL quantum efficiency of films reported in panel as a function of excitation power (532 nm, CW laser). Solid red squares are data taken 1 week prior to data marked with open squares, showing good sample stability. Reproduced with permission.^[82] Copyright 2016, American Chemical Society. g) Calculated hysteresis index as a function of $[M]/[Pb]$. Reproduced with permission.^[81] Copyright 2018, American Chemical Society.

of zwitterion or polyfunctional groups will be one of the future development directions.

3.2.4. Passivation by Wide Bandgap Materials

Materials with wider bandgaps than silicon such as SiO_2 , SiN and amorphous silicon are often used as surface passivation

layers, which are important tools on the way of improving the PCE of silicon-based solar cells. This classic and effective passivation strategy has already been applied to the PSCs.

Wide-bandgap inorganic oxide materials are preferred as passivators. Koushik et al. prepared Al_2O_3 layer on perovskite by atomic layer deposition (ALD),^[145] which was one of the commonly used passivation materials in silicon-based solar cells. They realized that holes need to be transported through

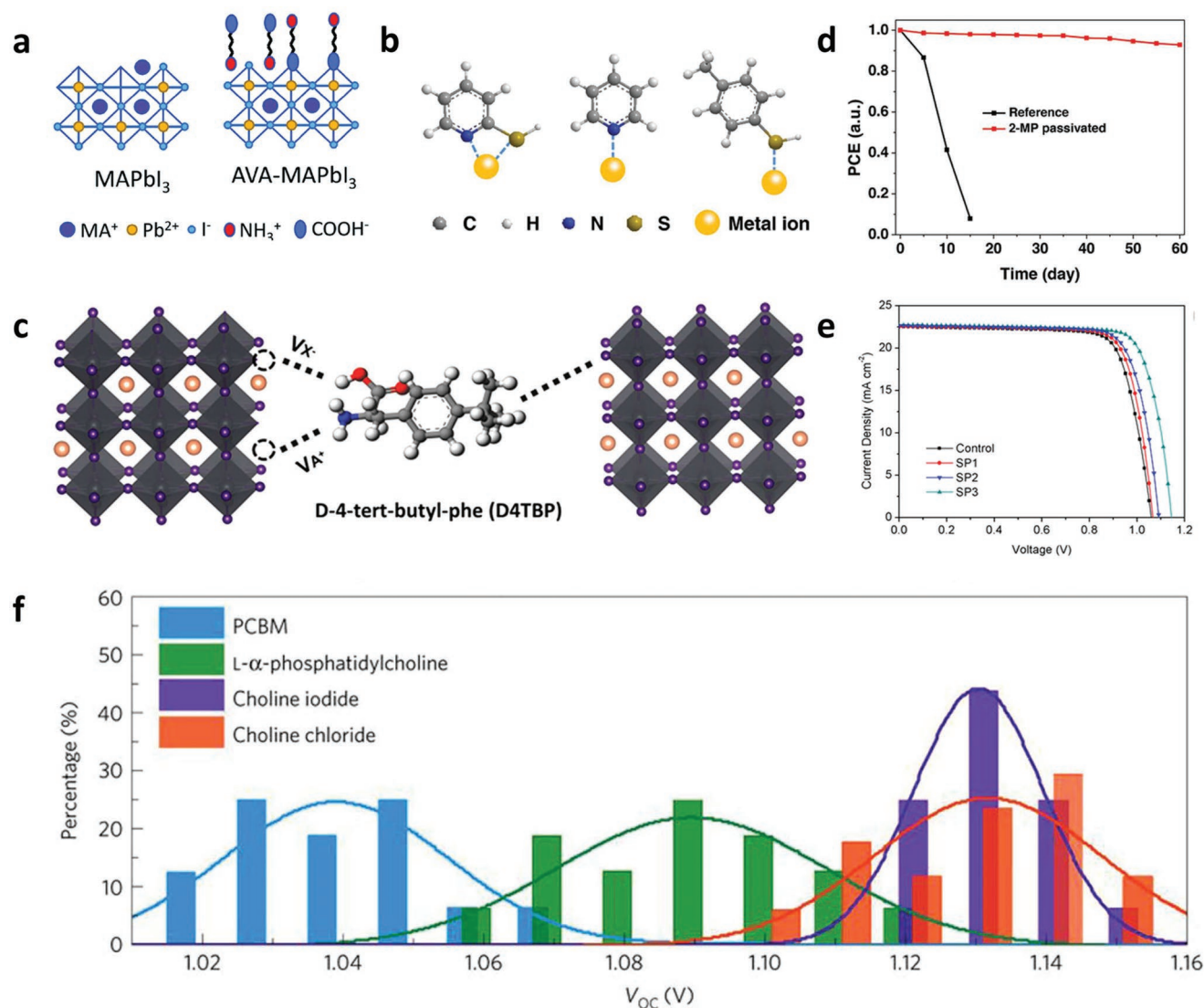


Figure 8. Passivation by polyfunctional Lewis base. a) Schematic representation of AVA passivation at lattice termination of MAPbI₃. Reproduced with permission.^[144] Copyright 2017, Macmillan. b) Passivating materials and schematic illustration of their interaction with metal ions (pyridine and *p*-toluenethiol, 2-MP). Reproduced with permission.^[85] Copyright 2019, Wiley-VCH. c) Schematic illustration of the origin of D4TBP passivation effect on different defect sites. Reproduced with permission.^[86] Copyright 2019, American Chemical Society. d) Stability test of the reference and 2-MP passivated devices stored in ambient air with a relative humidity of 60–70% at room temperature. Reproduced with permission.^[85] Copyright 2019, Wiley-VCH. e) J–V characteristic of the champion devices with a 0.09 cm² effective cell area under simulated AM1.5G solar illumination based of different perovskite absorbers. Reproduced with permission.^[89] Copyright 2019, Wiley-VCH. f) The statistics of V_{OC} distribution for devices with PCBM (blue), L-α-phosphatidylcholine (green), choline iodide (violet), and choline chloride (red). (Statistics from 16 samples on the same type of substrates for each passivation molecule.) The solid lines represent the Gauss distribution fitting for the statistic of V_{OC}. Reproduced with permission.^[87] Copyright 2017, Macmillan.

tunneling, so the Al₂O₃ passivation layer must be very thin, as thin as about 1 nm. In addition, Al₂O₃ can also act as an encapsulant.^[146] Bai et al. designed a new method to wrap the perovskite grains with oligomeric silica (OS) to form a core-shell structure, which not only passivated the surface and grain boundaries, but also fine-tuned the grain size (Figure 9a).^[100] This strategy allowed pure-MAPbI₃ devices to achieve a PCE of 21.5%, with a V_{OC} of 1.15 V.

Self-passivation effects formed by wide-bandgap perovskites are more universal. Wide-bandgap perovskites can usually adjust the energy levels between adjacent layers, using the

I-type band principle to hinder electron transport and reduce recombination (Figure 9c). Jung et al. stacked wide-bandgap perovskite onto the narrow bandgap perovskite surface by in-situ reaction of n-hexyl trimethyl ammonium bromide (HTAB) (Figure 9b).^[103] This layer not only greatly reduces the interfacial recombination, but also promotes the film formation of poly(3-hexylthiophene) (P3HT) through van der Waals force, making the device exhibit a PCE of 22.7%. There is no doubt that, similar to the above, most of the alkylamine salts mentioned in Section 3.2.2 also act as passivators by conversion to low-dimensional perovskites. Nevertheless, another

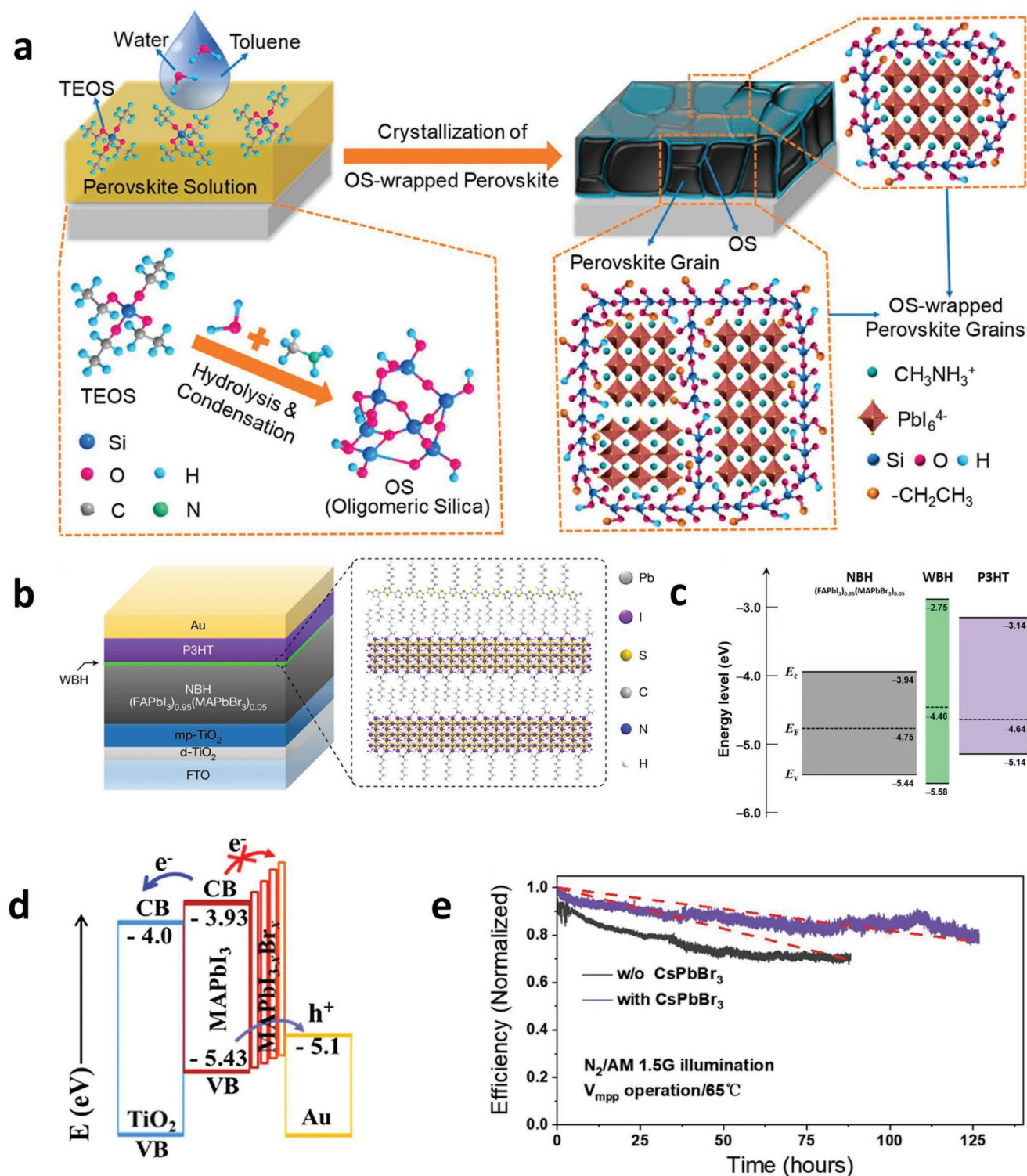


Figure 9. Passivation by wide bandgap materials. a) Schematic illustration showing the in situ formation process of an OS-wrapped perovskite thin film and the hypothesized perovskite-OS nanostructures. Reproduced with permission.^[100] Copyright 2019, American Chemical Society. b) Double-layered halide architecture of P3HT-based perovskite solar cells and schematic structure of the interface between the WBH (wide-bandgap halide) and P3HT.^[103] c) Energy-level diagram of NBH (narrow-bandgap halide), WBH, and P3HT. Reproduced with permission.^[103] Copyright 2019, Macmillan. d) Energy band alignment at the perovskite/Au interface for MAPbI₃/MAPbI_{3-x}Br_x (0 < x ≤ 1) device. Reproduced with permission.^[104] Copyright 2018, Elsevier. e) Steady-state output of PSCs under continuous MPP operation, AM1.5G illumination, and thermal stress for 125 h in nitrogen-filled glovebox (the devices are putting on a hot plate and the temperature of devices' surface are monitored as 65 °C). Reproduced with permission.^[105] Copyright 2019, Wiley-VCH.

self-passivated PSCs with gradually enlarged bandgap was invented by Ruan et al. through thermal evaporation of MABr.^[104] The gradually increasing conduction band minimum subtly blocks the passage of photo generated electrons, greatly reducing the interface recombination (Figure 9d).

Quantum dots (QDs) are another common type of passivators. CsPbBr₃ clusters have been shown to inhibit phase separation of multicomponent perovskites and improve device stability by increasing the activation energy of ion motion (Figure 9e).^[105] Further, CsPbBr_{1.85}I_{1.15} QDs form interface dipoles to increase the built-in electric field to enhance the ability to extract holes.^[106] Other quantum dot materials such as carbon quantum dots are also considered to have passivation effects.^[147,148]

3.2.5. Passivation by Hydrophobic Materials

Since the development of PSCs, its stability under water and oxygen has become a key issue to be solved. It is a good solution to add an insulating hydrophobic layer between the light absorbing layer and the charge transport layer, and the hydrophobic material can also reduce nonradiative recombination through specific functional groups like carbonyl,^[108–113] fluorine etc. while resisting moisture infiltration.^[62,114,149]

Due to their ultralong chain macromolecular structure, polymer materials like PVP,^[111] PMMA, and EVA can effectively prevent water molecules from contacting with perovskite.^[112,113] Wang et al. tested a range of common plastic materials including Teflon, PS plastic foam and fluoro-silane,^[108] and found that the thickness of this layer must be ultrathin due to its insulation. They proposed a model to explain the passivation mechanism of such materials that the insulating layer was selective for carrier tunneling, and only specific carriers could tunnel (Figure 10a). Moreover, the surface contact angle is significantly increased with the action of hydrophobic materials (Figure 10b).^[109] Kim et al. gave a new insight that the hygroscopic insulation layer also increases the humidity stability.^[110] Comparing polyethylene oxide (PEO) and polystyrene (PS), they concluded that PEO slowed the hydration of perovskite by absorbing water molecules before the perovskite did (Figure 10e), which allowed the device to remain for 15 days in the 88% humidity. Another interesting thing is that the addition of g-C₃N₄ or rubrene are believed to passivate lead clusters and promote hole extraction by embedding carrier shuttle paths.^[101,102,150]

Organic small molecules containing fluorine groups also help to improve the humidity stability of the device. Based on its extremely strong electronegativity, fluorine easily forms hydrogen bonds with water molecules, thus forming a protective barrier against moisture. Liu et al. introduced a 2D perovskite layer using pentafluoro-phenylethylammonium (FEA) as the A site cation over the 3D analogs, resulting in a device with PCE up to 22.6%.^[114] It was amazing that the device maintained 90% of the initial efficiency after 1000 h at 40% relative humidity in ambient condition with a prevalent humidity of 40% (Figure 10c). Ionic liquids containing fluorine groups have also been reported to improve the moisture resistance of devices. Related to its large size of ionic liquid molecule, 3-(3-aminopropyl)-1-methylimidazolium hexafluorophosphate

(APMimPF₆) was believed to resist water intrusion by forming a PF₆-based perovskite (Figure 10d).^[149] Another group used BMIMBF₄ to modify perovskite, exhibiting good long-term stability performance under full-spectrum sunlight and heat stressing at 70–75 °C (Figure 10f).^[62]

4. Passivation of Carrier Transport layer

In general, the structure of PSC devices can be divided into two main types according to the topography of the carrier transport layer (CTL): planar structure and mesoporous structure. The planar structure PSCs consists of two architectures: convention structure (cathode/electron transport layer (ETL)/perovskite/hole transport layer (HTL)/anode, n–i–p) and inverted one (anode/HTL/perovskite/ETL/cathode) (p–i–n). 25.2% of the champion PCE was created by mesoporous devices, while the highest PCE records achieved by the planar devices were 23.7% (n–i–p) and around 21% (p–i–n),^[151,152] respectively. As can be seen from these excellent devices, the choice of the transport layer material is crucial. The commonly used ETL materials are TiO₂, SnO₂, and PCBM, and the ordinary HTL materials are spiro-OMeTAD and PTAA.

It is believed that the valence band maximum (VBM) and semiconductor type of perovskite are related to film composition, defect type and substrate interface state,^[153] wherein the substrate is determined by the CTL below the perovskite layer. By studying the interface between adjacent layers of the device, the researchers found complex changes in the energy level at the interface, such as bending or dipole generation. Therefore, it is of importance for the under layer CTL to contain good film quality, achieve energetic alignment and good interface contact with the perovskite absorbing layer. In order to achieve the above objectives, a series of passivation measures for the CTL have been implemented. The details of these passivation strategies for device performance improvement are shown in Table 2.

4.1. Passivation by Doping in CTL

Compared with TiO₂, tin dioxide is a more ideal ETL material in n–i–p planar structure perovskite devices in view of its high light transmittance, high electron mobility and low temperature preparation characteristics.^[168] However, like TiO₂, the N-type conductivity of SnO₂ is derived from the native defects caused by oxygen vacancies, which in turn causes traps to capture electrons.^[169] The existence of internal defects may make the electron extraction capability of SnO₂ not match the hole extraction of HTL, which is considered to be the key factor for device hysteresis.

In order to reduce the hysteresis of the device, different kinds of additives are doped in the SnO₂ film formation process to passivate the defects. Many research groups have prepared SnO₂ thin films by different synthesis and deposition methods,^[170,171] but the obtained devices all have non-negligible hysteresis. Surprisingly, You et al. adopted SnO₂ nanoparticles as ETL for planar structure solar cells with nearly free hysteresis.^[172] In 2018, another team also found that there was almost no hysteresis in the film prepared by SnO₂ nanoparticles.^[160]

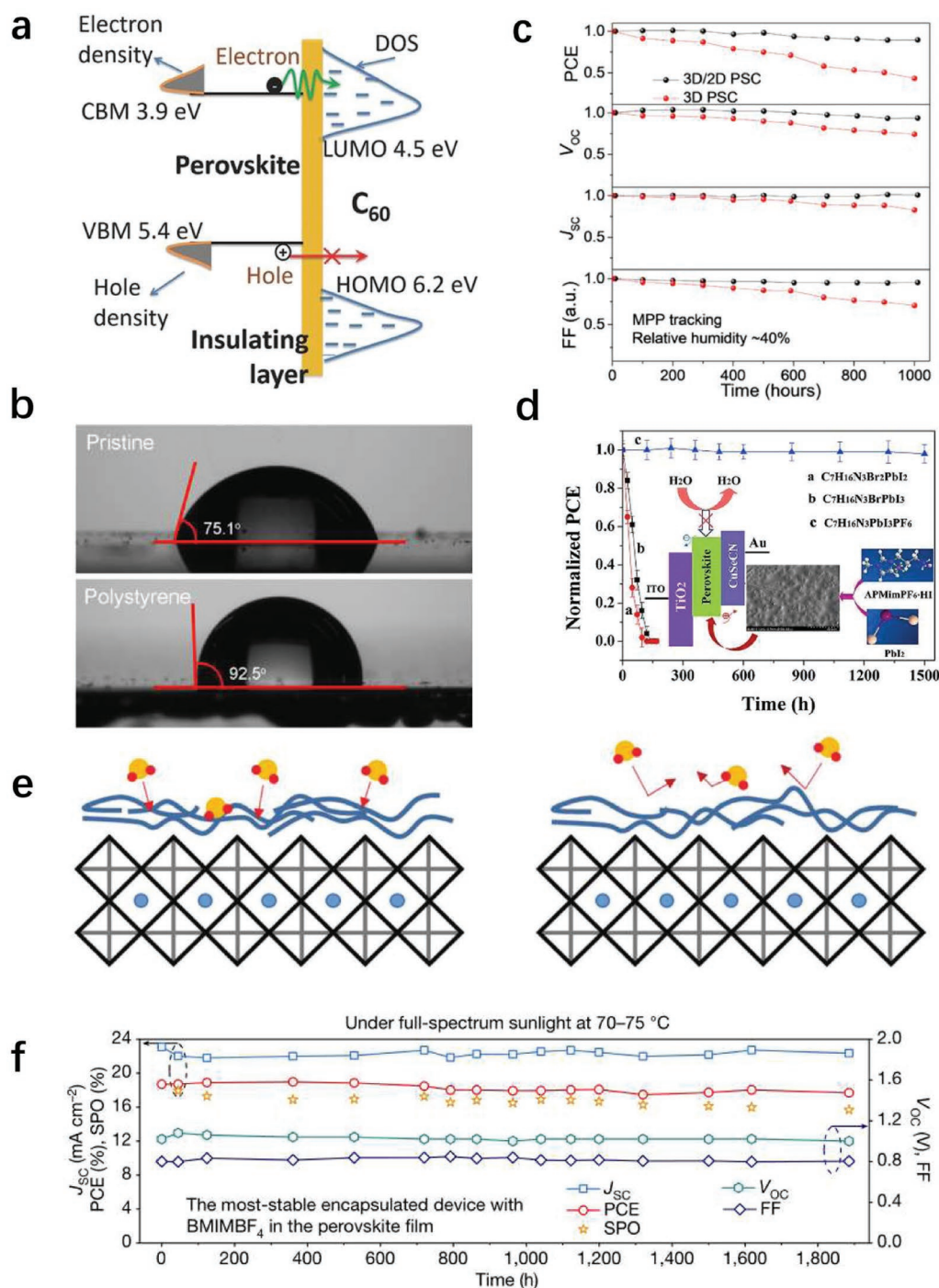


Figure 10. Passivation by hydrophobic materials. a) The energy diagram that illustrates the principle of suppressing surface charge recombination by the insulating layer. The insulating layer suppresses charge recombination by separating the excess electrons and holes in the electron transport layer and perovskite layer, respectively. Reproduced with permission.^[108] Copyright 2016, Wiley-VCH. b) Contact angle tests of the perovskite films without (top) and with (down) polystyrene layer. Reproduced with permission.^[109] Copyright 2018, American Chemical Society. c) Ambient atmosphere aging results of a 3D PSC and a 3D/2D PSC, with the relative humidity shown in the inset. Reproduced with permission.^[114] Copyright 2019, American Association for the Advancement of Science. d) Long-term stability test of photovoltaic performance of solar cells assembled with $C_7H_{16}N_3Br_2PbI_2$, $C_7H_{16}N_3BrPbI_3$ and $C_7H_{16}N_3PbI_3PF_6$ and schematic configuration of the solar cell structure. Reproduced with permission.^[149] Copyright 2019, Elsevier. e) Schematic illustration of a polymer thin film of polyethylene oxide (PEO) and polystyrene (PS) assembled on the perovskite structure. Reproduced with permission.^[110] Copyright 2018, The Royal Society of Chemistry. f) Long-term stability performance of the most stable device with BMIMBF₄ in the perovskite film, under full-spectrum sunlight and heat stressing at 70–75 °C. The device parameters were collected from the J - V curves for this device. The concentration of the BMIMBF₄ in the perovskite film is 0.3 mol% with respect to the lead atoms. Reproduced with permission.^[62] Copyright 2019, Macmillan.

Table 2. Summary of passivation strategies for charge transport layer.

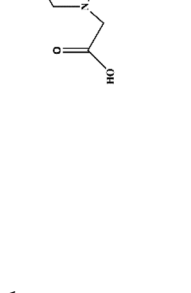
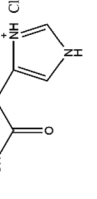
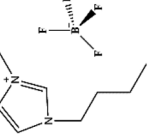
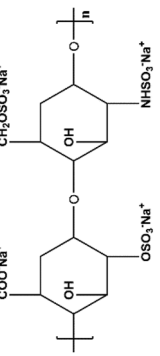
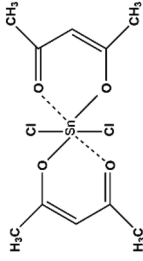
Passivator	Structure	Perovskite components	Passivation functional groups	Targeted defects	V_{oc} [V] (C/P)	J_{sc} [mA cm^{-2}] (C/P)	FF [%] (C/P)	PCE [%] (C/P)	Ref.
NH_4Cl		FAMAPb(I BrCl) $_3$	Cl^-	SnO_2	1.10/1.15	23.22/24.28	73.47/76.83	18.71/21.38	[49]
EDTA		CsFAPbI $_3$	$-\text{COOH}$	SnO_2	1.10/1.11	22.79/24.57	75.5/79.2	18.93/21.60	[154]
2,2,2-Trifluoroethanol		FAMAPb(I BrCl) $_3$	Fluorine substituent, O donor	SnO_2	1.10/1.12	23.12/23.91	75.5/78.0	19.17/20.92	[155]
KCl		$\text{Cs}_{0.05}\text{FA}_{0.8}\text{MA}_{0.14}\text{Pb}_{1.25}\text{Br}_{0.45}$	K^+	SnO_2	1.03/1.13	22.21/23.10	75.44/78.57	17.3/20.5	[156]
LiF		$\text{MAPbI}_{3-x}\text{Cl}_x$	F^-	SnO_2	1.08/1.15	20.40/21.62	71.1/73.7	15.6/18.33	[157]
ImAcHCl		(FAPbI $_3$) $_{0.95}$ (MAPbBr $_3$) $_{0.05}$	Ester bond	SnO_2	1.09/1.15	22.65/23.06	79/79	19.53/20.96	[158]
BMIMBF $_4$		MAPbI $_3$	BMIM $^+$, BF $_4^-$	TiO $_2$	1.08/1.12	22.31/22.75	75/77	18.14/19.62	[159]
Heparin sodium (HS)		MAPbI $_3$	$-\text{OSO}_3^-$	TiO $_2$	1.09/1.11	21.29/23.34	74.07/77.31	17.20/20.10	[142]
KOH		CsMAFAPb(I Br) $_3$	K^+	TiO $_2$	1.10/1.15	22.45/22.60	78/79	19.27/20.50	[160]
Cl $_2$		(FAPbI $_3$) $_{0.95}$ (MAPbI $_3$) $_{0.15}$	$-\text{C}=\text{O}$, $-\text{Cl}$	TiO $_2$	1.13	24.78	79	22.19	[161]

Table 2. Continued.

Passivator	Structure	Perovskite components	Passivation functional groups	Targeted defects	V_{oc} [V] (C/P)	J_{sc} [mA cm^{-2}] (C/P)	FF [%] (C/P)	PCE [%] (C/P)	Ref.
PVCz/PCBM		(CsRbFAMA)Pb(BrI) ₃	Aromatic ring, Fullerene	TiO ₂	1.07/1.12	23.2/23.2	78/79	19.3/20.4	[162]
ALD SnO ₂		(FAPbI ₃) _{0.85} (MAPbI ₃) _{0.15}	SnO ₂	TiO ₂ /SnO ₂	1.07/1.13	22.59/22.68	73/78	17.75/20.03	[163]
Monolayer graphene (MLG)		FAMAPb(I/BrCl) ₃	MLG	ZnO	1/1.12	20.63/22.71	67.3/77.9	13.82/19.81	[164]
Ethanolamine + MgO		(CsFAMA)Pb(BrI) ₃	—NH ₂	ZnO	1.09/1.12	22.86/23.86	73.52/78.91	18.31/21.08	[165]
PTAA		(FAPbI ₃) _{0.85} (MAPbBr ₃) _{0.15}	Aromatic ring	spiro-OMeTAD	1.10/1.10	20.98/21.32	67.13/75.70	15.49/17.75	[166]
Adamanty/ammonium iodide (ADAHl)		FAMAPb(I/Br) ₃	ADA, —NH ₂	spiro-OMeTAD	1.11/1.16	23.5/24.3	76.8/77.3	19.9/21.9	[167]

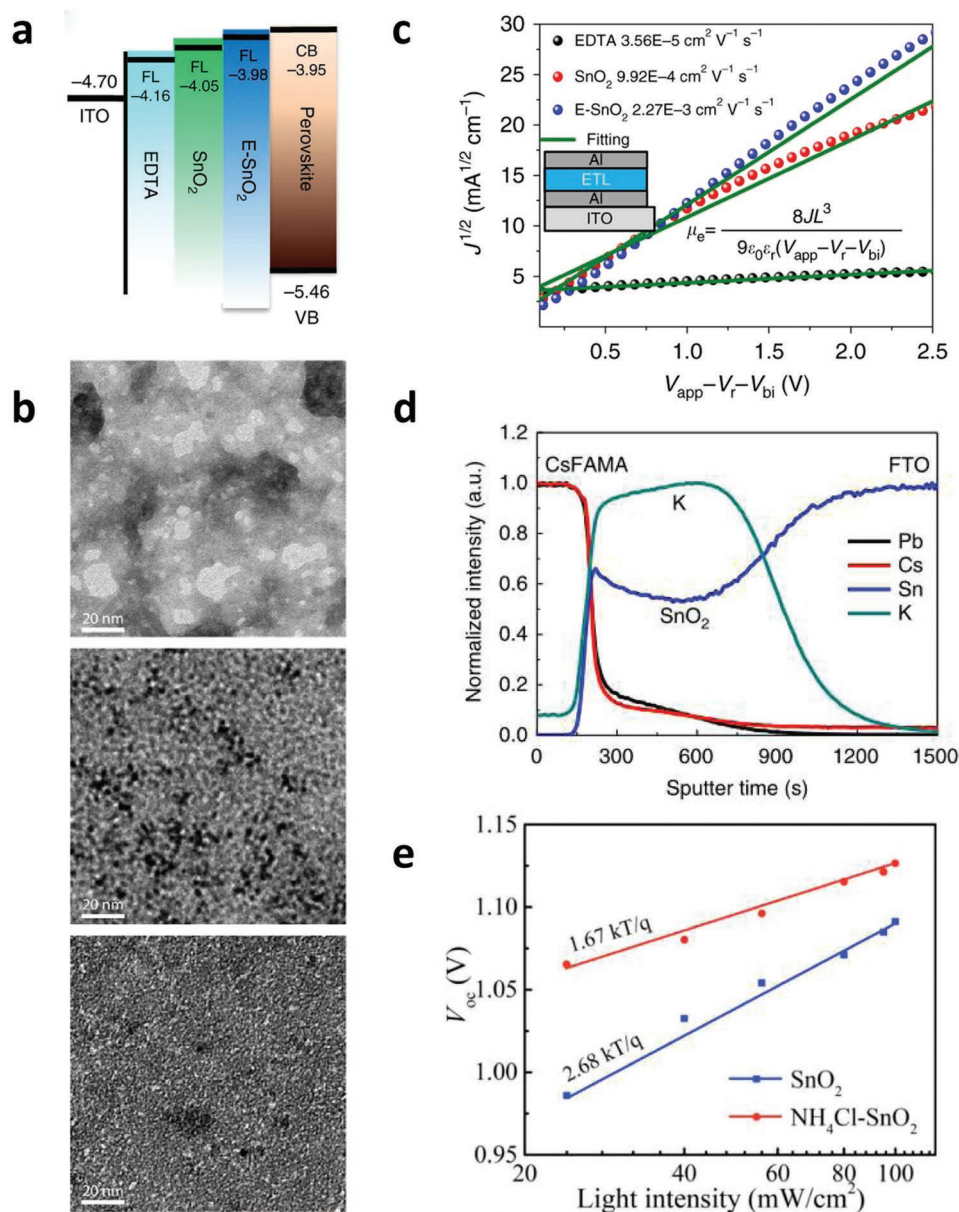


Figure 11. Passivation by doping in CTL. a) Schematic illustration of Fermi level of EDTA, SnO_2 , and E- SnO_2 relative to the conduction band of the perovskite layer. The Fermi level of EDTA, SnO_2 , and E- SnO_2 is measured by KPFM, and the conduction and valence bands of the perovskite materials are obtained from the previous report. Reproduced with permission.^[154] Copyright 2018, Macmillan. b) Bright-field (BF) TEM images of SnO_2 films deposited on a carbon support grid, and the corresponding selected area electron diffraction patterns (SAEDP) tin(II). From top to bottom, the solvent is bis(acetylacetonate), tin(IV) bis(acetylacetonate) dichloride, and tin(IV) bis(acetylacetonate) dibromide in turn. Reproduced with permission.^[161] Copyright 2019, The Royal Society of Chemistry. c) Electron mobility for EDTA, SnO_2 , and E- SnO_2 using the SCLC model, and the inset shows the device structure of ITO/Al/ETL/Al. Reproduced with permission.^[154] Copyright 2018, Macmillan. d) The normalized SIMS results of the CsFAMA perovskite deposited on Alfa- SnO_2 /FTO substrate. Reproduced with permission.^[160] Copyright 2018, Macmillan. e) V_{oc} versus light intensity for the PSCs using SnO_2 and $\text{NH}_4\text{Cl-SnO}_2$. Reproduced with permission.^[49] Copyright 2019, Wiley-VCH

After investigation by the scanning transmission electron microscopy and energy-dispersive X-ray spectroscopy (STEM-EDX), they believed that it was related to potassium ions in solution (Figure 11d). The test results of TOF-SIMS (time of flight secondary ion mass spectrometry) show that potassium ions are concentrated in the interface between SnO_2 and perovskite, which may have the same passivation effect as the incorporation of perovskite.

In addition, the introduction of chloride ions in SnO_2 by doping ammonium chloride in the precursor solution can also effectively eliminate hysteresis. According to the DFT calculation results, Cl^- and NH_4^+ ions make the system more stable at the interface,^[49] because it is more compatible with the crystal structure of SnO_2 than MA^+ and I^- in perovskite. The formation energy of the Pb-I antisite defect is also higher after the introduction of ammonium chloride. Reduced defects (as shown in

Figure 11e), more matched energy bands and increased electron mobility allowed the device to achieve a PCE of 21.38% with negligible hysteresis. Additionally, Liang et al. thought that NH_4^+ might react with OH^- in the SnO_2 solution, so that the Sn single bond was exposed and had a chance to bond with Cl^- .^[173]

Another effective way to passivate is to formulate the precursor solution by ethylene diamine tetraacetic acid (EDTA) complexation.^[154] Due to its strong chelation, EDTA has a good modification effect on SnO_2 . On the one hand, the modification of EDTA makes the electron mobility of SnO_2 greatly improved, which is increased by about three times, and the possible accumulation of charge is reduced (Figure 11c). On the other hand, the conduction band of SnO_2 is raised to be close to that of perovskite, which benefits the promotion of open-circuit voltage (Figure 11a).

Abuhelaia et al. compared three precursor solutions of SnO_2 and found that tin(IV) bis(acetylacetonate) dichloride (Cl_2) and tin(IV) bis(acetylacetonate) dibromide (Br_2) have significant passivation effects.^[161] The TEM image showed that the film with Cl_2 and Br_2 had a compact surface and uniform grain size (Figure 11b), and there was no pinhole that might become a leakage channel. This result may stem from that Sn^{4+} metal atom coordinate with two more mono-dentate halogen ligands in Cl_2/Br_2 than tin(II) bis(acetylacetonate). Moreover, Luan et al. also observed that oxygen plasma treatment of the SnO_2 film surface can significantly improve device performance, possibly attributed to changes in surface roughness and work function.^[155]

There are also many methods of passivation for other under layer materials, such as doping Li^+ to increase the electron mobility of TiO_2 ,^[174] or introducing $\text{Cu}^+/\text{Li}^+/\text{Mg}^{2+}$ to improve the conductivity of NiO .^[175,176] It is not discussed in detail here since the basic principle is similar to that of SnO_2 .

4.2. Passivation of Perovskite/CTL Interface

It is a common modification method to insert intermediate layer at the interface, which can reduce the nonradiative recombination of the under layer surface, block the transport of heterogeneous carriers or rivet the upper and lower layers. The conductive intermediate layer usually exhibits the effect of reducing defects and compensating for insufficient conductivity of the transport layer. For example, adding PCBM or graphene on the surface of TiO_2 or ZnO ^[177,164] and introducing KCl or KOH on the surface of SnO_2 (Figure 12a).^[160,156] Considering the compatibility of adjacent layers, it is troublesome to choose the right method to grow the interlayer. Okada et al. mixed poly(vinylcarbazole) (PVCz) with PCBM to coat onto TiO_2 ,^[162] where PVCz acted as a scaffold to fix PCBM, preventing PCBM from being washed away by the next spin-coating (Figure 12c). This method effectively reduced the electroluminescence intensity, indicating that the nonradiative recombination caused by interface defects is improved (Figure 12b). The work of Yang et al. proves that ionic liquids have similar effects.^[159] It has also been accounted to deposit dense SnO_2 as an intermediate passivation layer by ALD method.^[163] For the interfaces between perovskite and HTL, there is a report that the PTAA layer is inserted for passivation.^[166]

A typical representative of an insulated intermediate layer is LiF, and device bandgap alignment is shown in the Figure 12e.^[157] As a wide bandgap material, it can effectively block holes and promote electron transport. In order not to affect the morphology of the other layers, LiF is deposited by thermal evaporation under high vacuum. It is found that the addition of LiF layer could significantly increase the V_{OC} of the device, but the thickness of the layer would affect the series resistance according to the Nyquist plot (Figure 12f). Further comparing the effects of PbI_2 and LiTFSi , the author denoted that fluoride ions in LiF played a key role in suppressing surface trap states.

Another type of interlayer connects the transport layer with perovskites by acting like a rivet. Cao et al. introduced a method for effectively passivating the surface of ZnO film.^[165] As shown in Figure 12d, a thin layer of MgO prevents ZnO from contacting with perovskites to inhibit proton extraction, while protonated ethanolamine (EA) anchors perovskite through amine groups to reduce interface instability and contact barrier. The introduction of EA made the PCE of PSCs with ZnO as the ETL for the first time break to 21%. For TiO_2 or SnO_2 , there are also attempts to anchor with polyfunctional groups by biopolymer heparin sodium or ImAcHCl .^[142,158]

5. Summary and Outlook

In this review, we systematically expounded the passivation strategies of internal, surface, and interface in perovskite solar cell devices in recent years, with particular emphasis on the mechanism of various methods. In general, passivators upgrade the device performance in the following ways:

- 1) Fill in vacancies or fix interstitial atoms with specific functional groups or other chemical interactions carried by the passivators.
- 2) Anchor the adjacent layer by means of intermolecular forces and optimize interface contact.
- 3) Form energy level steps to improve carrier extraction capability.
- 4) Act as a barrier to block internal minority carrier transport and ion migration as well as external water molecules and gold atoms penetration.

Through the above functions, passivation has become an indispensable optimization strategy for PSCs to be more efficient and stable. And it is exploited by devices that create world records in PCE. Although a lot of work on the mechanism of different molecular passivation has been done, there is still a lack of consensus on the specific effects. We recommend that a systematic study should be conducted to give a convincing answer to the influence of the location, quantity or other factors of a particular functional group or other chemical bond, which may greatly help to screen out the really suitable passivators in the future. In addition, more characterization methods and theoretical calculations should also be involved.

Cooperative passivation by multifunctionality or multiple passivators has proven to be an effective way to improve the effect of passivation. However, this strategy is currently limited to play a role in the perovskite layer without considering the carrier transport layer. In fact, there are a large number of

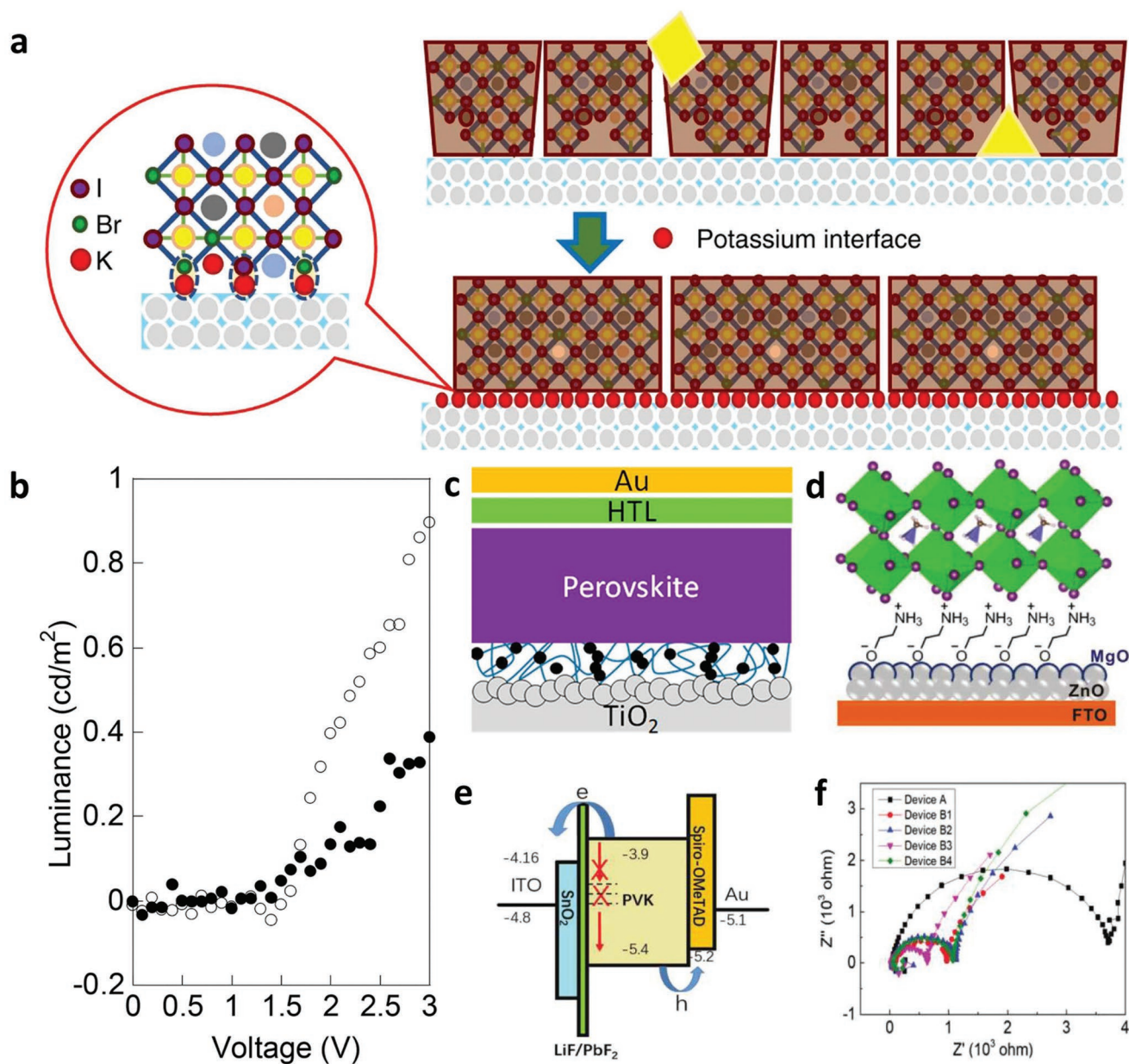


Figure 12. Passivation by adding an intermediate layer. a) A schematic of the perovskite growth process on different substrates. Reproduced with permission.^[160] Copyright 2018, Macmillan. b) Luminescence intensity from the PSCs with the PVCz/PCBM modification on the TiO₂ layer (●) and without (○).^[162] c) Schematic diagram of device with PVCz anchoring PCBM. Reproduced with permission.^[162] Copyright 2019, American Chemical Society. d) Schematic illustration of a planar PSC device modified with ZnO-MgO-EA⁺. Reproduced with permission.^[165] Copyright 2018, Wiley-VCH. e) Energy level structure of the PSC.^[157] f) The Nyquist plot of the PSC under 1 Sun. Reproduced with permission.^[157] Copyright 2018, The Royal Society of Chemistry.

defects in the interior and the interface of the transport layer. Therefore, the comprehensive passivation of overall device should be the development trend to improve the performance of PSCs in the future.

In view of the theoretical low defect density and high carrier mobility,^[178] single crystal (SC) perovskites are considered to have the opportunity to break the efficiency limits of current polycrystalline perovskite devices. Up to now, the PCE of SC-PSCs has exceeded 21%.^[179] According to the history of the

development of monocrystalline silicon solar cells, we have no doubt that further performance enhancement of SC-PSCs must rely on targeted surface passivation. The relevant experience and research methods of silicon surface passivation are worth learning.

Finally, we believe that through the in-depth understanding of the passivation mechanism and the application of comprehensive passivation measures, the PCE and stability of PSCs will make a huge leap in the near future.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

defect passivation, high-efficiency, perovskites, solar cells

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- [1] A. Kojima, K. Teshima, Y. Shirai, T. Miyasaka, *J. Am. Chem. Soc.* **2009**, *131*, 6050.
- [2] NREL Efficiency Chart Vol. 2019, <https://www.nrel.gov/pv/assets/pdfs/best-research-cell-efficiencies.20190802.pdf> (accessed: August 2019).
- [3] M. Cai, Y. Wu, H. Chen, X. Yang, Y. Qiang, L. Han, *Adv. Sci.* **2017**, *4*, 1600269.
- [4] Z. Li, Y. Zhao, X. Wang, Y. Sun, Z. Zhao, Y. Li, H. Zhou, Q. Chen, *Joule* **2018**, *2*, 1559.
- [5] Z. Song, C. L. McElvany, A. B. Phillips, I. Celik, P. W. Krantz, S. C. Watthage, G. K. Liyanage, D. Apul, M. J. Heben, *Energy Environ. Sci.* **2017**, *10*, 1297.
- [6] L. Meng, J. You, Y. Yang, *Nat. Commun.* **2018**, *9*, 5265.
- [7] L. M. Pazos-Outón, T. P. Xiao, E. Yablonovitch, *J. Phys. Chem. Lett.* **2018**, *9*, 1703.
- [8] G. Grancini, C. Roldán-Carmona, I. Zimmermann, E. Mosconi, X. Lee, D. Martineau, S. Nabey, F. Oswald, F. De Angelis, M. Graetzel, M. K. Nazeeruddin, *Nat. Commun.* **2017**, *8*, 15684.
- [9] H. Kim, J. Seo, N. Park, *ChemSusChem* **2016**, *9*, 2528.
- [10] V. Rofati, E. Mosconi, A. Listorti, S. Colella, G. Gigli, F. de Angelis, *Nano Lett.* **2014**, *14*, 2168.
- [11] J. Carrillo, A. Guerrero, S. Rahimnejad, O. Almora, I. Zarazua, E. Mas-Marza, J. Bisquert, G. Garcia-Belmonte, *Adv. Energy Mater.* **2016**, *6*, 1502246.
- [12] B. Vermang, Y. Ren, O. D. Gargand, C. Frisk, J. Joel, P. Salomé, J. Borme, S. Sadewasser, C. Björkman, M. Edoff, *IEEE J. Photovoltaics* **2016**, *6*, 332.
- [13] J. Kim, S. H. Lee, J. H. Lee, K. H. Hong, *J. Phys. Chem. Lett.* **2014**, *5*, 1312.
- [14] W.-J. Yin, T. Shi, Y. Yan, *Appl. Phys. Lett.* **2014**, *104*, 063903.
- [15] M. L. Agiorgousis, Y. Sun, H. Zeng, S. Zhang, *J. Am. Chem. Soc.* **2014**, *136*, 14570.
- [16] A. Walsh, D. O. Scanlon, S. Chen, X. G. Gong, S. H. Wei, *Angew. Chem., Int. Ed.* **2015**, *54*, 1791.
- [17] C. Eames, J. M. Frost, P. R. F. Barnes, B. C. O'Regan, A. Walsh, M. S. Islam, *Nat. Commun.* **2015**, *6*, 7497.
- [18] A. Buin, P. Pietsch, J. Xu, O. Voznyy, A. H. Ip, R. Comin, E. H. Sargent, *Nano Lett.* **2014**, *14*, 6281.
- [19] J. Xu, A. Buin, A. H. Ip, W. Li, O. Voznyy, R. Comin, M. Yuan, S. Jeon, Z. Ning, J. J. McDowell, P. Kanjanaboos, J. Sun, X. Lan, L. N. Quan, D. H. Kim, I. G. Hill, P. Maksymovych, E. H. Sargent, *Nat. Commun.* **2015**, *6*, 7081.
- [20] A. Buin, R. Comin, J. Xu, A. H. Ip, E. H. Sargent, *Chem. Mater.* **2015**, *27*, 4405.
- [21] X. Wu, M. T. Trinh, D. Niesner, H. Zhu, Z. Norman, J. S. Owen, O. Yaffe, B. J. Kudsich, X.-Y. Zhu, *J. Am. Chem. Soc.* **2015**, *137*, 2089.
- [22] J. M. Ball, A. Petrozza, *Nat. Energy* **2016**, *1*, 16149.
- [23] J. Nelson, *The Physics of Solar Cells*, Imperial College Press, London **2003**.
- [24] B. Chen, P. N. Rudd, S. Yang, Y. Yuan, J. Huang, *Chem. Soc. Rev.* **2019**, *48*, 3842.
- [25] K. Tvingstedt, O. Malinkiewicz, A. Baumann, C. Deibel, H. J. Snaith, V. Dyakonov, H. J. Bolink, *Sci. Rep.* **2015**, *4*, 6071.
- [26] W. Tress, N. Marinova, O. Inganäs, M. K. Nazeeruddin, S. M. Zakeeruddin, M. Graetzel, *Adv. Energy Mater.* **2015**, *5*, 1400812.
- [27] F. Staub, H. Hempel, J.-C. Hebig, J. Mock, U. W. Paetzold, U. Rau, T. Unold, T. Kirchartz, *Phys. Rev. Appl.* **2016**, *6*, 044017.
- [28] W. Shockley, W. T. Read, *Phys. Rev.* **1952**, *87*, 835.
- [29] R. N. Hall, *Phys. Rev.* **1952**, *87*, 387.
- [30] J. Huang, Y. Yuan, Y. Shao, Y. Yan, *Nat. Rev. Mater.* **2017**, *2*, 17042.
- [31] M. A. Green, A. Baillie, H. J. Snaith, *Nat. Photonics* **2014**, *8*, 506.
- [32] J. C. Baena, A. Abate, M. Saliba, W. Tress, T. J. Jacobsson, M. Graetzel, A. Hagfeldt, *Energy Environ. Sci.* **2017**, *10*, 710.
- [33] S. D. Stranks, P. K. Nayak, W. Zhang, T. Stergiopoulos, H. J. Snaith, *Angew. Chem., Int. Ed.* **2015**, *54*, 3240.
- [34] R. S. Bonilla, B. Hoex, P. Hamer, P. R. Wilshaw, *Phys. Status Solidi A* **2017**, *214*, 1700293.
- [35] T. S. Sherkar, C. Momblona, L. Gil-Escrig, J. Ávila, M. Sessolo, H. J. Bolink, L. J. A. Koster, *ACS Energy Lett.* **2017**, *2*, 1214.
- [36] W. Tress, N. Marinova, T. Moehl, S. M. Zakeeruddin, M. K. Nazeeruddin, M. Graetzel, *Energy Environ. Sci.* **2015**, *8*, 995.
- [37] J. M. Azpiroz, E. Mosconi, J. Bisquert, F. De Angelis, *Energy Environ. Sci.* **2015**, *8*, 2118.
- [38] D. H. Kang, N. G. Park, *Adv. Mater.* **2019**, *31*, 1805214.
- [39] Q. Jiang, Y. Zhao, X. Zhang, X. Yang, Y. Chen, Z. Chu, Q. Ye, X. Li, Z. Yin, J. You, *Nat. Photonics* **2019**, *13*, 460.
- [40] P. W. Liang, C. C. Chueh, S. T. Williams, A. Jen, *Adv. Energy Mater.* **2015**, *5*, 1402321.
- [41] T. Niu, J. Lu, M. Tang, D. Barrit, D. Smilgies, Z. Yang, J. Li, Y. Fan, T. Luo, I. McCulloch, A. Amassian, S. Liu, K. Zhao, *Energy Environ. Sci.* **2018**, *11*, 3358.
- [42] N. K. Noel, A. Abate, S. D. Stranks, E. S. Parrott, V. M. Burlakov, A. Goriely, H. J. Snaith, *ACS Nano* **2014**, *8*, 9815.
- [43] Y. Shao, Z. Xiao, C. Bi, Y. Yuan, J. Huang, *Nat. Commun.* **2014**, *5*, 5784.
- [44] Q. Chen, H. Zhou, T. Song, S. Luo, Z. Hong, H. Duan, L. Dou, Y. Liu, Y. Yang, *Nano Lett.* **2014**, *14*, 4158.
- [45] E. Edri, S. Kirmayer, S. Mukhopadhyay, K. Gartsman, G. Hodes, D. Cahen, *Nat. Commun.* **2014**, *5*, 3461.
- [46] L. K. Ono, Y. Qi, J. Phys. Chem. Lett. **2016**, *7*, 4764.
- [47] C. Bi, X. Zheng, B. Chen, H. Wei, J. Huang, *ACS Energy Lett.* **2017**, *2*, 1400.
- [48] N. Li, S. Tao, Y. Chen, X. Niu, C. K. Onwudinanti, C. Hu, Z. Qiu, Z. Xu, G. Zheng, L. Wang, Y. Zhang, L. Li, H. Liu, Y. Lun, J. Hong, X. Wang, Y. Liu, H. Xie, Y. Gao, Y. Bai, S. Yang, G. Brocks, Q. Chen, H. Zhou, *Nat. Energy* **2019**, *4*, 408.
- [49] Z. Liu, K. Deng, J. Hu, L. Li, *Angew. Chem.* **2019**, *131*, 2.
- [50] D. Son, S. Kim, J. Seo, S. Lee, H. Shin, D. Lee, N. Park, *J. Am. Chem. Soc.* **2018**, *140*, 1358.
- [51] M. A. Jalebi, Z. A. Garmaroudi, S. Cacovich, C. Stavrakas, B. Philippe, J. M. Richter, M. Alsari, E. P. Booker, E. M. Hutter, A. J. Pearson, S. Lilliu, T. J. Savenije, H. Rensmo, G. Divitini, C. Ducati, R. H. Friend, S. D. Stranks, *Nature* **2018**, *555*, 497.
- [52] M. A. Jalebi, Z. A. Garmaroudi, A. J. Pearson, G. Divitini, S. Cacovich, B. Philippe, H. Rensmo, C. Ducati, R. H. Friend, S. D. Stranks, *ACS Energy Lett.* **2018**, *3*, 2671.

- [53] F. Zheng, W. Chen, T. Bu, K. P. Ghiggino, F. Huang, Y. Cheng, P. Tapping, T. W. Kee, B. Jia, X. Wen, *Adv. Energy Mater.* **2019**, 9, 1901016.
- [54] Y. Guo, F. Zhao, J. Tao, J. Jiang, J. Zhang, J. Yang, Z. Hu, J. Chu, *ChemSusChem* **2019**, 12, 983.
- [55] S. Yuan, Y. Cai, S. Yang, H. Zhao, F. Qian, Y. Han, J. Sun, Z. Liu, S. Liu, *Sol. RRL* **2019**, 3, 1900220.
- [56] X. Gong, L. Guan, H. Pan, Q. Sun, X. Zhao, H. Li, H. Pan, Y. Shen, Y. Shao, L. Sun, Z. Cui, L. Ding, M. Wang, *Adv. Funct. Mater.* **2018**, 28, 1804286.
- [57] L. Wang, H. Zhou, J. Hu, B. Huang, M. Sun, B. Dong, G. Zheng, Y. Huang, Y. Chen, L. Li, Z. Xu, N. Li, Z. Liu, Q. Chen, L. Sun, C. Yan, *Science* **2019**, 363, 265.
- [58] J. Lu, S. Chen, Q. Zheng, *Sci. China: Chem.* **2019**, 62, 1044.
- [59] M. Chen, M. Ju, H. F. Garces, A. D. Carl, L. K. Ono, Z. Hawash, Y. Zhang, T. Shen, Y. Qi, R. L. Grimm, D. Pacifici, X. Zeng, Y. Zhou, N. P. Padture, *Nat. Commun.* **2019**, 10, 16.
- [60] M. Kim, G. Kim, T. Lee, I. Choi, H. Choi, Y. Jo, Y. Yoon, J. Kim, J. Lee, D. Huh, H. Lee, S. Kwak, J. Kim, D. Kim, *Joule* **2019**, 3, 1464.
- [61] S. Yuan, F. Qian, S. Yang, Y. Cai, Q. Wang, J. Sun, Z. Liu, S. Liu, *Adv. Funct. Mater.* **2019**, 1807850.
- [62] S. Bai, P. Da, C. Li, Z. Wang, Z. Yuan, Fan Fu, M. Kaweck, X. Liu, N. Sakai, J. Wang, S. Huettner, S. Buecheler, M. Fahlman, F. Gao, H. J. Snaith, *Nature* **2019**, 571, 245.
- [63] M. M. Tavakoli, M. Saliba, P. Yadav, P. Holzhey, A. Hagfeldt, S. M. Zakeeruddin, M. Grätzel, *Adv. Energy Mater.* **2019**, 9, 1802646.
- [64] Q. Jiang, Z. Chu, P. Wang, X. Yang, H. Liu, Y. Wang, Z. Yin, J. Wu, X. Zhang, J. You, *Adv. Mater.* **2017**, 29, 1703852.
- [65] Y. Zhao, Q. Li, W. Zhou, Y. Hou, Y. Zhao, R. Fu, D. Yu, X. Liu, Q. Zhao, *Sol. RRL* **2018**, 8, 9815.
- [66] D. Son, J. Lee, Y. Choi, I. Jang, S. Lee, P. Yoo, H. Shin, N. Ahn, M. Choi, D. Kim, N. Park, *Nat. Energy* **2016**, 1, 16081.
- [67] E. A. Alharbi, A. Y. Alyamani, D. J. Kubicki, A. R. Uhl, B. J. Walder, A. Q. Alanazi, J. Luo, A. B. Caminal, A. Albadi, H. Albrithen, M. H. Alotaibi, J. E. Moser, S. M. Zakeeruddin, F. Giordano, L. Emsley, M. Grätzel, *Nat. Commun.* **2019**, 10, 3008.
- [68] Z. Wang, Q. Lin, F. P. Chmiel, N. Sakai, L. M. Herz, H. J. Snaith, *Nat. Energy* **2017**, 2, 17135.
- [69] E. Jokar, C. Chien, A. Fathi, M. Rameez, Y. Chang, E. W. Diau, *Energy Environ. Sci.* **2018**, 11, 2353.
- [70] Y. Cho, A. Soufiani, J. Yun, J. Kim, D. Lee, J. Seidel, X. Deng, M. A. Green, S. Huang, A. W. Y. Ho-Baillie, *Adv. Energy Mater.* **2018**, 8, 1703392.
- [71] M. Jung, T. Shin, J. Seo, G. Kim, S. I. Seok, *Energy Environ. Sci.* **2018**, 11, 2188.
- [72] T. Zhao, C.-C. Chueh, Q. Chen, A. Rajagopal, A. K.-Y. Jen, *ACS Energy Lett.* **2016**, 1, 757.
- [73] M. M. Tavakoli, P. Yadav, D. Prochowicz, M. Sponseller, A. Osherov, V. Bulović, J. Kong, *Adv. Energy Mater.* **2019**, 9, 1803587.
- [74] N. Li, Z. Zhu, Q. Dong, J. Li, Z. Yang, C. Chueh, A. Jen, L. Wang, *Adv. Mater. Interfaces* **2017**, 4, 1700598.
- [75] F. Wang, W. Geng, Y. Zhou, H. Fang, C. Tong, M. A. Loi, L. Liu, N. Zhao, *Adv. Mater.* **2016**, 28, 9986.
- [76] S. Zhao, J. Xie, G. Cheng, Y. Xiang, H. Zhu, W. Guo, H. Wang, M. Qin, X. Lu, J. Qu, J. Wang, J. Xu, K. Yan, *Small* **2018**, 14, 1803350.
- [77] A. Abate, M. Saliba, D. J. Hollman, S. D. Stranks, K. Wojciechowski, R. Avolio, G. Grancini, A. Petrozza, H. J. Snaith, *Nano Lett.* **2014**, 14, 3247.
- [78] M. Tavakoli, D. Bi, L. Pan, A. Hagfeldt, S. Zakeeruddin, M. Grätzel, *Adv. Energy Mater.* **2018**, 8, 1800275.
- [79] G. Yang, P. Qin, G. Fang, G. Li, *Sol. RRL* **2018**, 2, 1800055.
- [80] Y. Zhang, G. Grancini, Z. Fei, E. Shirzadi, X. Liu, E. Oveisi, F. F. Tirani, R. Scopelliti, Y. Feng, M. K. Nazeeruddin, P. J. Dyson, *Nano Energy* **2019**, 58, 105.
- [81] S. G. Kim, J. Z. Chen, J. Y. Seo, D. H. Kang, N. G. Park, *ACS Appl. Mater. Interfaces* **2018**, 10, 25372.
- [82] D. W. deQuilettes, S. Koch, S. Burke, R. K. Paranj, A. J. Shropshire, M. E. Ziffer, D. S. Ginger, *ACS Energy Lett.* **2016**, 1, 438.
- [83] B. Wang, F. Wu, S. Bi, J. Zhou, J. Wang, X. Leng, D. Zhang, R. Meng, B. Xue, C. Zong, Linna Zhu, Y. Zhang, H. Zhou, *J. Mater. Chem. A* **2019**, <https://doi.org/10.1039/C9TA01947B>.
- [84] J. Yang, S. Xiong, T. Qu, Y. Zhang, X. He, X. Guo, Q. Zhao, S. Braun, J. Chen, J. Xu, Y. Li, X. Liu, C. Duan, J. Tang, M. Fahlman, Q. Bao, *ACS Appl. Mater. Interfaces* **2019**, 11, 13491.
- [85] H. Zhang, Y. Wu, C. Shen, E. Li, C. Yan, W. Zhang, H. Tian, L. Han, W. Zhu, *Adv. Energy Mater.* **2019**, 9, 1803573.
- [86] S. Yang, J. Dai, Z. Yu, Y. Shao, Y. Zhou, X. Xiao, X. C. Zeng, J. Huang, *J. Am. Chem. Soc.* **2019**, 141, 5781.
- [87] X. P. Zheng, B. Chen, J. Dai, Y. J. Fang, Y. Bai, Y. Z. Lin, H. T. Wei, X. C. Zeng, J. S. Huang, *Nat. Energy* **2017**, 2, 17102.
- [88] X. Zheng, Y. Deng, B. Chen, H. Wei, X. Xiao, Y. Fang, Y. Lin, Z. Yu, Y. Liu, Q. Wang, J. Huang, *Adv. Mater.* **2018**, 30, 1803428.
- [89] T. Wu, Y. Wang, X. Li, Y. Wu, X. Meng, D. Cui, X. Yang, L. Han, *Adv. Energy Mater.* **2019**, 9, 1803766.
- [90] Y. Guo, J. Ma, H. Lei, F. Yao, B. Li, L. Xiong, G. Fang, *J. Mater. Chem. A* **2018**, 6, 5919.
- [91] X. Du, R. Qiu, T. Zou, X. Chen, H. Chen, H. Zhou, *Adv. Mater. Interfaces* **2019**, 6, 1900413.
- [92] Q. Tai, X. Guo, G. Tang, P. You, T. Ng, D. Shen, J. Cao, C. Liu, N. Wang, Y. Zhu, C. Lee, F. Yan, *Angew. Chem., Int. Ed.* **2019**, 58, 806.
- [93] W. Chen, Y. Wang, G. Pang, C. Koh, A. Djurišić, Y. Wu, B. Tu, F. Liu, R. Chen, H. Woo, X. Guo, Z. He, *Adv. Funct. Mater.* **2019**, 1808855.
- [94] P. Qin, G. Yang, Z. Ren, S. H. Cheung, S. K. So, Li Chen, J. Hao, J. Hou, G. Li, *Adv. Mater.* **2018**, 30, 1706126.
- [95] L. Gao, S. Huang, L. Chen, X. Li, B. Ding, S. Huang, G. Yang, *Sol. RRL* **2018**, 2, 1800088.
- [96] J. Yang, C. Liu, C. Cai, X. Hu, Z. Huang, X. Duan, X. Meng, Z. Yuan, L. Tan, Y. Chen, *Adv. Energy Mater.* **2019**, 9, 1900198.
- [97] J. Tian, Q. Xue, X. Tang, Y. Chen, N. Li, Z. Hu, T. Shi, X. Wang, F. Huang, C. Brabec, H. Yip, Y. Cao, *Adv. Mater.* **2019**, 31, 1901152.
- [98] Z. Chen, M. Liu, Z. Li, T. Shi, Y. Yang, H. Yip, Y. Cao, *iScience* **2018**, 9, 337.
- [99] Z. Wang, A. Pradhan, M. Kamarudin, M. Pandey, S. Pandey, P. Zhang, C. Ng, A. Tripathi, T. Ma, S. Hayase, *ACS Appl. Mater. Interfaces* **2019**, 11, 10012.
- [100] Y. Bai, Y. Lin, L. Ren, X. Shi, E. Strounina, Y. Deng, Q. Wang, Y. Fang, X. Zheng, Y. Lin, Z. Chen, Y. Du, L. Wang, J. Huang, *ACS Energy Lett.* **2019**, 4, 1231.
- [101] J. Liao, W. Wu, J. Zhong, Y. Jiang, L. Wang, D. Kuang, *J. Mater. Chem. A* **2019**, 7, 9025.
- [102] L. Z. Wang, M. Li, C. Zhang, Q. Ye, K. Hu, D. Lu, P. Fang, L. Liao, *Adv. Funct. Mater.* **2018**, 28, 1705875.
- [103] E. H. Jung, N. J. Jeon, E. Y. Park, C. S. Moon, T. J. Shin, T. Yang, J. H. Noh, J. Seo, *Nature* **2019**, 567, 511.
- [104] W. Zhou, S. Chen, Y. Zhao, Q. Li, Y. Zhao, R. Fu, D. Yu, P. Gao, Q. Zhao, *Adv. Funct. Mater.* **2019**, 29, 1809180.
- [105] W. Ruan, Z. Zhang, Y. Hu, F. Bai, T. Qiu, S. Zhang, *Appl. Surf. Sci.* **2019**, 465, 420.
- [106] S. Akin, Y. Altintas, E. Mutlugun, S. Sonmezoglu, *Nano Energy* **2019**, 60, 557.
- [107] J. J. Yoo, S. Wieghold, M. C. Sponseller, M. R. Chua, S. N. Bertram, N. T. P. Hartono, J. S. Tresback, E. C. Hansen, J. P. C. Baena, V. Bulović, T. Buonassisi, S. S. Shin, M. G. Bawendi, *Energy Environ. Sci.* **2019**, 12, 2192.
- [108] Q. Wang, Q. Dong, T. Li, A. Gruverman, J. Huang, *Adv. Mater.* **2016**, 28, 6734.
- [109] M. Li, X. Yan, Z. Kang, Y. Huan, Y. Li, R. Zhang, Y. Zhang, *ACS Appl. Mater. Interfaces* **2018**, 10, 18787.

- [110] M. Kim, S. G. Motti, R. Sorrentino, A. Petrozza, *Energy Environ. Sci.* **2018**, *11*, 2609.
- [111] B. Li, Y. Zhang, L. Fu, T. Yu, S. Zhou, L. Zhang, L. Yin, *Nat. Commun.* **2018**, *9*, 1076.
- [112] J. Peng, J. I. Khan, W. Liu, E. Ugur, T. Duong, Y. Wu, H. Shen, K. Wang, H. Dang, E. Aydin, X. Yang, Y. Wan, K. J. Weber, K. R. Catchpole, F. Laquai, S. D. Wolf, T. P. White, *Adv. Energy Mater.* **2018**, *8*, 1801208.
- [113] Z. Huang, X. Hu, C. Liu, X. Meng, Z. Huang, J. Yang, X. Duan, J. Long, Z. Zhao, L. Tan, Y. Song, Y. Chen, *Adv. Funct. Mater.* **2019**, 1902629.
- [114] Y. Liu, S. Akin, L. Pan, R. Uchida, N. Arora, J. V. Milić, A. Hinderhofer, F. Schreiber, A. R. Uhl, S. M. Zakeeruddin, A. Hagfeldt, M. I. Dar, M. Grätzel, *Sci. Adv.* **2019**, *5*, eaaw2543.
- [115] M. L. Agiorgousis, Y.-Y. Sun, H. Zeng, S. Zhang, *J. Am. Chem. Soc.* **2014**, *136*, 14570.
- [116] S. Yun, X. Zhou, J. Even, A. Hagfeldt, *Angew. Chem., Int. Ed.* **2017**, *56*, 15806.
- [117] Y.-T. Hsieh, Q. Han, C. Jiang, T.-B. Song, H. Chen, L. Meng, H. Zhou, Y. Yang, *Adv. Energy Mater.* **2016**, *6*, 1502386.
- [118] D. J. Kubicki, D. Prochowicz, A. Hofstetter, S. M. Zakeeruddin, M. Grätzel, L. Emsley, *J. Am. Chem. Soc.* **2018**, *140*, 7232.
- [119] D.-Y. Son, J.-W. Lee, Y. J. Choi, I.-H. Jang, S. Lee, P. J. Yoo, H. Shin, N. Ahn, M. Choi, D. Kim, N.-G. Park, *Nat. Energy* **2016**, *1*, 16081.
- [120] F. Y. Zhang, B. Yang, X. Mao, R. X. Yang, L. Jiang, Y. J. Li, J. Xiong, Y. Yang, R. X. He, W. Q. Deng, K. L. Han, *ACS Appl. Mater. Interfaces* **2017**, *9*, 14827.
- [121] C. M. Sutter-Fella, Y. B. Li, M. Amani, J. W. Ager, F. M. Toma, E. Yablonovitch, I. D. Sharp, A. Javey, *Nano Lett.* **2016**, *16*, 800.
- [122] J. He, T. Chen, *J. Mater. Chem. A* **2015**, *3*, 18514.
- [123] G. Y. Kim, S. H. Oh, B. P. Nguyen, W. Jo, B. J. Kim, D. G. Lee, H. S. Jung, *J. Phys. Chem. Lett.* **2015**, *6*, 2355.
- [124] Z. Wang, Y. Zhou, S. Pang, Z. Xiao, J. Zhang, W. Chai, H. Xu, Z. Liu, N. P. Padture, G. Cui, *Chem. Mater.* **2015**, *27*, 7149.
- [125] F. Xie, C.-C. Chen, Y. Wu, X. Li, M. Cai, X. Liu, X. Yang, L. Han, *Energy Environ. Sci.* **2017**, *10*, 1942.
- [126] N. Yantara, F. Yanan, C. Shi, H. A. Dewi, P. P. Boix, S. G. Mhaisalkar, N. Mathews, *Chem. Mater.* **2015**, *27*, 2309.
- [127] V. L. Pool, A. Gold-Parker, M. D. McGehee, M. F. Toney, *Chem. Mater.* **2015**, *27*, 7240.
- [128] W. J. Yin, H. Y. Chen, T. T. Shi, S. H. Wei, Y. F. Yan, *Adv. Electron. Mater.* **2015**, *1*, 1500044.
- [129] W. Ke, C. Xiao, C. Wang, B. Saparov, H.-S. Duan, D. Zhao, Z. Xiao, P. Schulz, S. P. Harvey, W. Liao, W. Meng, Y. Yu, A. J. Cimaroli, C.-S. Jiang, K. Zhu, M. Al-Jassim, G. Fang, D. B. Mitzi, Y. Yan, *Adv. Mater.* **2016**, *28*, 5214.
- [130] T. Supasai, N. Rujisamphan, K. Ullrich, A. Chemseddine, T. Dittrich, *Appl. Phys. Lett.* **2013**, *103*, 183906.
- [131] L. L. Wang, C. McCleese, A. Kovalsky, Y. X. Zhao, C. Burda, *J. Am. Chem. Soc.* **2014**, *136*, 12205.
- [132] M. C. Shih, S. S. Li, C. H. Hsieh, Y. C. Wang, H. D. Yang, Y. P. Chiu, C. S. Chang, C. W. Chen, *Nano Lett.* **2017**, *17*, 1154.
- [133] A. G. Aberle, *Prog. Photovoltaics* **2000**, *8*, 473.
- [134] Z. Hawash, S. R. Raga, D. Y. Son, L. K. Ono, N. G. Park, Y. B. Qi, *J. Phys. Chem. Lett.* **2017**, *8*, 3947.
- [135] M. Z. Long, T. K. Zhang, H. Y. Zhu, G. X. Li, F. Wang, W. Y. Guo, Y. Chai, W. Chen, Q. Li, K. S. Wong, J. B. Xu, K. Y. Yan, *Nano Energy* **2017**, *33*, 485.
- [136] Y. Chen, Y. Sun, J. Peng, W. Zhang, X. Su, K. Zheng, T. Pullerits, Z. Liang, *Adv. Energy Mater.* **2017**, *7*, 1700162.
- [137] M. Tavakoli, P. Yadav, D. Prochowicz, M. Sponseller, A. Osharov, V. Bulović, J. Kong, *Adv. Energy Mater.* **2019**, *9*, 1803587.
- [138] Y. Hu, J. Schlipf, M. Wussler, M. L. Petrus, W. Jaegermann, T. Bein, P. M. Buschbaum, P. Docampo, *ACS Nano* **2016**, *10*, 5999.
- [139] D. Lee, J. Yun, J. Kim, A. Soufiani, S. Chen, Y. Cho, X. Deng, J. Seidel, S. Lim, S. Huang, A. W. Y. Ho-Baillie, *ACS Energy Lett.* **2018**, *3*, 647.
- [140] T. Niu, J. Lu, R. Munir, J. Li, D. Barrit, X. Zhang, H. Hu, Z. Yang, A. Amassian, K. Zhao, S. Liu, *Adv. Mater.* **2018**, *30*, 1706576.
- [141] X. Fang, J. N. Ding, N. Y. Yuan, P. Sun, M. H. Lv, G. Q. Ding, C. Zhu, *Phys. Chem. Chem. Phys.* **2017**, *19*, 6057.
- [142] S. You, H. Wang, S. Bi, J. Zhou, L. Qin, X. Qiu, Z. Zhao, Y. Xu, Y. Zhang, X. Shi, H. Zhou, Z. Tang, *Adv. Mater.* **2018**, *30*, 1706924.
- [143] X. Fang, J. Ding, N. Yuan, P. Sun, M. Lv, G. Ding, C. Zhu, *Phys. Chem. Chem. Phys.* **2017**, *19*, 6057.
- [144] C. Lin, F. D. Rossi, J. Kim, J. Baker, J. Ngiam, B. Xu, S. Pont, N. Aristidou, S. A. Haque, T. Watson, M. A. McLachlan, J. R. Durrant, *J. Mater. Chem. A* **2019**, *7*, 3006.
- [145] D. Koushik, W. Verhees, Y. Kuang, S. Veenstra, D. Zhang, M. A. Verheijen, M. Creatore, R. E. I. Schropp, *Energy Environ. Sci.* **2017**, *10*, 91.
- [146] E. Y. Choi, J. Kim, S. Lim, E. Han, A. W. Y. Ho-Baillie, N. Park, *Sol. Energy Mater. Sol. Cells* **2018**, *188*, 37.
- [147] Q. Guo, F. Yuan, B. Zhang, S. Zhou, J. Zhang, Y. Bai, L. Fan, T. Hayat, A. Alsaedi, Z. Tan, *Nanoscale* **2019**, *11*, 115.
- [148] D. Benetti, E. Jokar, C. Yu, A. Fathi, H. Zhao, A. Vomiero, E. Diau, F. Rosei, *Nano Energy* **2019**, *62*, 781.
- [149] J. Wang, X. Ye, Y. Wang, Z. Wang, W. Wong, C. Li, *Electrochim. Acta* **2019**, *303*, 133.
- [150] P. Qin, J. Zhang, G. Yang, X. Yu, G. Li, *J. Mater. Chem. A* **2019**, *7*, 1824.
- [151] M. A. Green, Y. Hishikawa, E. D. Dunlop, D. H. Levi, J. Hohl-Ebinger, M. Yoshita, A. W. Y. Ho-Baillie, *Prog. Photovoltaics* **2019**, *27*, 3.
- [152] D. Luo, W. Yang, Z. Wang, A. Sadhanala, Q. Hu, R. Su, R. Shivanna, G. F. Trindade, J. F. Watts, Z. Xu, T. Liu, K. Chen, F. Ye, P. Wu, L. Zhao, J. Wu, Y. Tu, Y. Zhang, X. Yang, W. Zhang, R. H. Friend, Q. Gong, H. J. Snaith, R. Zhu, *Science* **2018**, *360*, 1442.
- [153] S. Wang, T. Sakurai, W. Wen, Y. Qi, *Adv. Mater. Interfaces* **2018**, *5*, 1800260.
- [154] D. Yang, R. Yang, K. Wang, C. Wu, X. Zhu, J. Feng, X. Ren, G. Fang, S. Priya, S. Liu, *Nat. Commun.* **2018**, *9*, 3239.
- [155] Y. Luan, X. Yi, P. Mao, Y. Wei, J. Zhuang, N. Chen, T. Lin, C. Li, J. Wang, *iScience* **2019**, *16*, 433.
- [156] X. Liu, Y. Zhang, L. Shi, Z. Liu, J. Huang, J. Yun, Y. Zeng, A. Pu, K. Sun, Z. Hameiri, J. Stride, J. Seidel, M. Green, X. Hao, *Adv. Energy Mater.* **2018**, *8*, 1800138.
- [157] S. Yuan, J. Wang, K. Yang, P. Wang, X. Zhang, Y. Zhan, L. Zheng, *Nanoscale* **2018**, *10*, 18909.
- [158] J. Chen, X. Zhao, S. Kim, N.-G. Park, *Adv. Mater.* **2019**, *31*, 1902902.
- [159] D. Yang, X. Zhou, R. Yang, Z. Yang, W. Yu, X. Wang, C. Li, S. Liu, R. P. H. Chang, *Energy Environ. Sci.* **2016**, *9*, 3071.
- [160] T. Bu, J. Li, F. Zheng, W. Chen, X. Wen, Z. Ku, Y. Peng, J. Zhong, Y. Cheng, F. Huang, *Nat. Commun.* **2018**, *9*, 4609.
- [161] M. Abuhelaiqa, S. Paek, Y. Lee, K. Cho, S. Heo, E. Oveisi, A. Huckaba, H. Kanda, H. Kim, Y. Zhang, R. Humphry-Baker, S. Kinge, A. Asiri, M. Nazeeruddin, *Energy Environ. Sci.* **2019**, *12*, 1910.
- [162] W. Okada, T. Suga, K. i Oyaizu, H. Segawa, H. Nishide, *ACS Appl. Energy Mater.* **2019**, *2*, 2848.
- [163] Y. Lee, S. Lee, G. Seo, S. Paek, K. Cho, A. Huckaba, M. Calizzi, D. Choi, J. Park, D. Lee, H. Lee, A. Asiri, M. Nazeeruddin, *Adv. Sci.* **2018**, *5*, 1800130.
- [164] M. Tavakoli, R. Tavakoli, P. Yadav, *J. Mater. Chem. A* **2019**, *7*, 679.

- [165] J. Cao, B. Wu, R. Chen, Y. Wu, Y. Hui, B. Mao, N. Zheng, *Adv. Mater.* **2018**, *30*, 1705596.
- [166] F. Hou, B. Shi, T. Li, C. Xin, Y. Ding, C. Wei, G. Wang, Y. Li, Y. Zhao, X. Zhang, *ACS Appl. Mater. Interfaces* **2019**, *11*, 25218.
- [167] M. Tavakoli, W. Tress, J. Milić, D. Kubicki, L. Emsley, M. Gratzel, *Energy Environ. Sci.* **2018**, *11*, 3310.
- [168] M. Park, J. Kim, H. Son, C. Lee, S. Jang, M. Ko, *Nano Energy* **2016**, *26*, 208.
- [169] K. Godinho, A. Walsh, G. Watson, *J. Phys. Chem. C* **2009**, *113*, 439.
- [170] C. Wang, L. Guan, D. Zhao, Y. Yu, C. Grice, Z. Song, R. Awni, J. Chen, J. Wang, X. Zhao, Y. Yan, *ACS Energy Lett.* **2017**, *2*, 2118.
- [171] E. Anaraki, A. Kermanpur, M. Mayer, L. Steier, T. Ahmed, S. Turren-Cruz, J. Seo, J. Luo, S. Zakeeruddin, W. Tress, T. Edvinsson, M. Gratzel, A. Hagfeldt, J. Correa-Baena, *ACS Energy Lett.* **2018**, *3*, 773.
- [172] Q. Jiang, L. Zhang, H. Wang, X. Yang, J. Meng, H. Liu, Z. Yin, J. Wu, X. Zhang, J. You, *Nat. Energy* **2017**, *2*, 16177.
- [173] J. Liang, Z. Chen, G. Yang, H. Wang, F. Ye, C. Tao, G. Fang, *ACS Appl. Mater. Interfaces* **2019**, *11*, 23152.
- [174] N. Arora, M. Dar, M. Abdi-Jalebi, F. Giordano, N. Pellet, G. Jacopin, R. H. Friend, S. M. Zakeeruddin, M. Graezel, *Nano Lett.* **2016**, *16*, 7155.
- [175] W. Chen, F. Liu, X. Feng, A. Djurišić, W. Chan, Z. He, *Adv. Energy Mater.* **2017**, *7*, 1700722.
- [176] W. Chen, Y. Wu, Y. Yue, J. Liu, W. Zhang, X. Yang, H. Chen, E. Bi, I. Ashraful, M. Gratzel, L. Han, *Science* **2015**, *350*, 944.
- [177] Y. Li, Y. Zhao, Q. Chen, Y. Yang, Y. Liu, Z. Hong, Z. Liu, Y. Hsieh, L. Meng, Y. Li, Y. Yang, *J. Am. Chem. Soc.* **2015**, *137*, 15540.
- [178] D. Shi, V. Adinolfi, R. Comin, M. Yuan, E. i Alarousu, A. Buin, Y. Chen, S. Hoogland, A. Rothenberger, K. Katsiev, Y. Losovyj, X. Zhang, P. Dowben, O. Mohammed, E. Sargent, O. Bakr, *Science* **2015**, *347*, 519.
- [179] Z. Chen, B. Turedi, A. Alsalloum, C. Yang, X. Zheng, I. Gereige, A. AlSaggaf, O. Mohammed, O. Bakr, *ACS Energy Lett.* **2019**, *4*, 1258.